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LIGHTNING, THUNDER AND LIGHTNING CONDUCTORS.

BY
GERALD MOLLOY, D.D., D.Sc.

ILLUSTRATED.

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*** START OF THE PROJECT GUTENBERG EBOOK LIGHTNING,
THUNDER AND LIGHTNING CONDUCTORS ***

LIGHTNING, THUNDER
AND
LIGHTNING CONDUCTORS.

*WITH AN APPENDIX ON THE RECENT CONTROVERSY ON
LIGHTNING CONDUCTORS.*

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LECTURE I.

LIGHTNING AND THUNDER.

The electricity produced by an ordinary electric machine exhibits, under certain conditions, phenomena which bear a striking resemblance to the phenomena attendant on lightning. In both cases there is a flash of light; in both there is a report, which, in the case of lightning, we call thunder; and, in both cases, intense heat is developed, which is capable of setting fire to combustible bodies. Further, the spark from an electric machine travels through space with extraordinary rapidity, and so does a flash of lightning; the spark follows a zig-zag course, and so does a flash of lightning; the spark moves silently and harmlessly through metal rods and stout wires, while it forces its way, with destructive effect, through bad conductors, and it is so, too, with a flash of lightning. Lastly, the electricity of a machine is capable of giving a severe shock to the human body; and we know that lightning gives a shock so severe as usually to cause immediate death. For these reasons it was long conjectured by scientific men that lightning is, in its nature, identical with electricity; and that it differs from the electricity of our machines only in this, that it exists in a more powerful and destructive form.

Identity of Lightning and Electricity.—But it was reserved for the celebrated Benjamin Franklin to demonstrate the truth of this conjecture by direct experiment. He first conceived the idea of drawing electricity from a thundercloud in the same way as it is drawn from the conductor of an electric machine. For this purpose he proposed to place a kind of sentry-box on the summit of a lofty tower, and to erect, on the sentry-box, a metal rod, projecting twenty or thirty feet upward into the air, pointed at the end, and having no electrical communication with the earth. He predicted that when a thundercloud would pass over the tower, the metal rod would become charged with electricity, and that an observer, stationed in the sentry-box, might draw from it, at pleasure, a succession of electric sparks.

With the magnanimity of a really great man, Franklin published this project to the world; being more solicitous to extend the domain of science

by new discoveries, than to secure for himself the glory of having made them. The project was set forth in a letter to Mr. Collinson, of London, which bears date July 29, 1750, and which, in the course of a year or two, was translated into the principal languages of Europe. Two years later the experiment suggested by Franklin was made by Monsieur Dalibard, a wealthy man of science, at his villa near Marly-la-Ville, a few miles from Paris. In the middle of an elevated plain Monsieur Dalibard erected an iron rod, forty feet in length, one inch in diameter, and ending above in a sharp steel point. The iron rod rested on an insulating support, and was kept in position by means of silk cords.

In the absence of Monsieur Dalibard, who was called by business to Paris, this apparatus was watched by an old dragoon, named Coiffier; and on the afternoon of the tenth of May, 1752, he drew sparks from the lower end of the rod at the time that a thundercloud was passing over the neighborhood. Conscious of the importance that would be attached to this phenomenon, the old dragoon summoned, in all haste, the prior of Marly to come and witness it. The prior came without delay, and he was followed by some of the principal inhabitants of the village. In the presence of the little group, thus gathered together, the experiment was repeated—electric sparks were again drawn, in rapid succession, from the iron rod; the prediction of Franklin was fulfilled to the letter; and the identity of lightning and electricity was, for the first time, demonstrated to the world.

Franklin's Experiment.—Meanwhile Franklin had been waiting, with impatience, for the completion of the tower of Christchurch, in Philadelphia, on which he intended to make the experiment himself. He even collected money, it is said, to hasten on the building. But, notwithstanding his exertions, the progress of the tower was slow; and his active mind, which could ill brook delay, hit upon another expedient, remarkable alike for its simplicity and for its complete success. He constructed a boy's kite, using, however, a silk pocket-handkerchief, instead of paper, that it might not be damaged by rain. To the top of the kite he attached a pointed iron wire about a foot long, and he provided a roll of hempen twine, which he knew to be a conductor of electricity, for flying it. This was the apparatus with which he proposed to explore the nature of a thundercloud.

The thundercloud came late in the afternoon of the fourth of July, 1752, and Franklin sallied out with his kite, accompanied by his son, and taking with him a common door-key and a Leyden jar. The kite was soon high in air, and the philosopher awaited the result of his experiment, standing, with his son, under the lee of a cowshed, partly to protect himself from the rain that was coming, and partly, it is said, to shield himself from the ridicule of passers-by, who, having no sympathy with his philosophical speculations, might be inclined to regard him as a lunatic. To guard against the danger of receiving a flash of lightning through his body, he held the kite by means of a silk ribbon, which was tied to the door-key, the door-key being itself attached to the lower end of the hempen string.

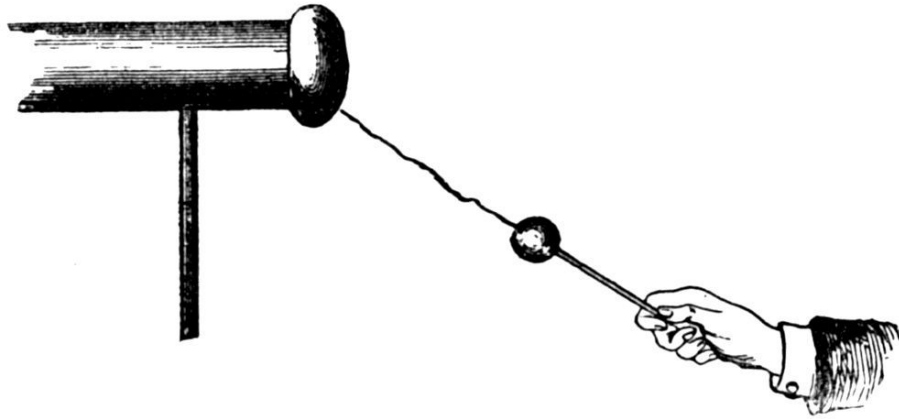
A flash of lightning soon came from the cloud, and a second, and a third; but no sign of electricity could be observed in the kite, or the hempen cord, or the key. Franklin was almost beginning to despair of success, when suddenly he noticed that the little fibres of the cord began to bristle up, just as they would if it were placed near an electric machine in action. He presented the door-key to the knob of the Leyden jar, and a spark passed between them. Presently a shower began to fall; the cord, wetted by the rain, became a better conductor than it had been before, and sparks came more freely. With these sparks he now charged the Leyden jar, and found, to his intense delight, that he could exhibit all the phenomena of electricity by means of the lightning he had drawn from the clouds.

In the following year a similar experiment, with even more striking results, was carried out, in France, by de Romas. Though it is said he had no knowledge of what Franklin had done in America, he, too, used a kite; and, with a view of making the string a better conductor, he interlaced with it a thin copper wire. Then, flying his kite in the ordinary way, when it had risen to a height of about 550 feet, he drew sparks from it which, we are told, were upwards of nine feet long, and emitted a sound like the report of a pistol.

Fatal Experiment of Richman.—There can be no doubt that experiments of this kind, made with the electricity of a thundercloud, were extremely dangerous; and this was soon proved by a fatal accident. Professor Richman, of St. Petersburg, had erected on the roof of his house a pointed iron rod, the lower end of which passed into a glass vessel, intended, as we are informed, to measure the strength of the charge which

he expected to receive from the clouds. On the sixth of August, 1753, observing the approach of a thunderstorm, he hastened to his apparatus; and as he stood near it, with his head bent down, to watch the effect, a flash of lightning passed through his body and killed him on the spot. This catastrophe served to fix public attention on the danger of such experiments, and gave occasion to the saying of Voltaire: “There are some great lords whom we should always approach with extreme precaution, and lightning is one of them.”^[1] From this time the practice of making experiments directly with the lightning of the clouds seems to have been, by common consent, abandoned.

Immediate Cause of Lightning.—And now, having set before you some of the most memorable experiments by which the identity of lightning and electricity has been demonstrated, I will try to give you a clear conception regarding the immediate cause of lightning, so far as the subject is understood at the present day by scientific men. You know that there are two kinds of electricity, which are called *positive* and *negative*; and that each of them repels electricity of the same kind as itself, while it attracts electricity of the opposite kind. Now, every thundercloud is charged with electricity of one kind or the other, positive or negative; and, as it hovers over the earth, it develops, by what is called *induction*, or influence, electricity of the opposite kind in that part of the earth which is immediately under it. Thus we have two bodies—the cloud and the earth—charged with opposite kinds of electricity, and separated by a stratum of the atmosphere. The two opposite electricities powerfully attract each other; but for a time they are prevented from rushing together by the intervening stratum of air, which is a non-conductor of electricity, and acts as a barrier between them. As the electricity, however, continues to accumulate, the attraction becomes stronger and stronger, until at length it is able to overcome the resistance of this barrier; a violent disruptive discharge then takes place between the cloud and the earth, and the flash of lightning is the consequence of the discharge.



THE ELECTRIC SPARK; A TYPE OF A FLASH OF LIGHTNING.

The whole phenomenon may be illustrated, on a small scale, by means of this electric machine of Carré's which you see before you. When my assistant turns the handle of the machine negative electricity is developed in that large brass cylinder, which in our experiment will represent the thundercloud. At a distance of five or six inches from the cylinder I hold a brass ball, which is in electrical communication with the earth through my body. The electrified brass cylinder acts by induction, or influence on the brass ball, and develops in it, as well as in my body, a charge of positive electricity. Now, the positive electricity of the ball and the negative electricity of the cylinder are mutually attracting each other, but the intervening stratum of air offers a resistance which prevents a discharge from taking place. My assistant, however, continues to work the machine; the two opposite electricities rapidly accumulate on the cylinder and the ball; at length their mutual attraction is strong enough to overcome the resistance interposed between them; a disruptive discharge follows, and at the same moment a spark is seen to pass, accompanied by a sharp snapping report.

This spark is a miniature flash of lightning; and the snapping report is a diminutive peal of thunder. Furthermore, at the moment the spark passes you may observe a slight convulsive movement in my hand and wrist. This convulsive movement represents, on a small scale, the violent shock, generally fatal to life, which is produced by a flash of lightning when it passes through the body.

I can continue to take sparks from the conductor as long as the machine is worked; and it is interesting to observe that these sparks follow an

irregular zig-zag course, just as lightning does. The reason is the same in both cases: a discharge between two electrified bodies takes place along the line of least resistance; and, owing to the varying condition of the atmosphere, as well as of the minute particles of matter floating in it, the line of least resistance is almost always a zig-zag line.

What a Flash of Lightning is.—Lightning, then, may be conceived as an electrical discharge, sudden and violent in its character, which takes place, through the atmosphere, between two bodies highly charged with opposite kinds of electricity. Sometimes this electrical discharge passes, as I have said, between a cloud and the earth; sometimes it passes between one cloud and another; sometimes, on a smaller scale, it takes place, between the great mass of a cloud and its outlying fragments.

But, if you ask me in what the discharge itself consists, I am utterly unable to tell you. It is usual to speak and write on this subject as if electricity were a material substance, a very subtle fluid, and as if, at the moment the discharge takes place, this fluid passes like a rapid stream, from the body that is positively electrified to the body that is negatively electrified. But we must always remember that this is only a conventional mode of expression, intended chiefly to assist our conceptions, and to help us to talk about the phenomena. It does not even profess to represent the objective truth. All that we know for certain is this: that immediately before the discharge the two bodies are highly electrified with opposite kinds of electricity; and, that immediately after the discharge, they are found to have returned to their ordinary condition, or, at least, to have become less highly electrified than they were before.

The flash of light that accompanies an electric discharge is often supposed to be the electricity itself, passing from one body to the other. But it is not; it is simply an effect produced by the discharge. Heat is generated by the expenditure of electrical energy, in overcoming the resistance offered by the atmosphere; and this heat is so intense, that it produces a brilliant incandescence along the path of the discharge. When a spark appears, for example, between the conductor of the machine and this brass ball, it can be shown, by very satisfactory evidence, that minute particles of these solid bodies are first converted into vapor, and then made to glow with intense heat. The gases, too, of which the air is composed, and the solid particles floating in the air, are likewise raised to incandescence. So, too, with

lightning; the flash of light is due to the intense heat generated by the electrical discharge, and owes its character to the composition and the density of the atmosphere through which the discharge passes.

Duration of a Flash of Lightning.—How long does a flash of lightning last? You are aware, I dare say, that when an impression of light is made on the eye, the impression remains for a sensible interval of time, not less than the tenth of a second, after the source of light has been extinguished or removed. Hence we continue, in fact, to see the light, for at least the tenth of a second, after the light has ceased. Now, if you reflect how brief is the moment for which a flash of lightning is visible, and if you deduct the tenth of a second from that brief moment, you will see, at once, that the period of its actual duration must be very short indeed.

The exact duration of a flash of lightning is a question on which no settled opinion has yet been accepted generally by scientific men. Indeed, the most widely different statements have been made on the subject, quite recently, by the highest authorities, each speaking apparently with unhesitating confidence. Thus, for example, Professor Mascart describes an experiment, which he says was made by Wheatstone, and which showed that a flash of lightning lasts for less than *one-thousandth* of a second;^[2] Professor Everett describes the same experiment, without saying by whom it was made, and gives, as the result, that “the duration of the illumination produced by lightning is certainly less than the *ten-thousandth* of a second;”^[3] Professor Tyndall, in his own picturesque way, tells us that “a flash of lightning cleaves a cloud, appearing and disappearing in less than the *hundred-thousandth* of a second;”^[4] and according to Professor Tait, of Edinburgh, “Wheatstone has shown that lightning certainly lasts less than the *millionth* of a second.”^[5]

Experiments of Professor Rood.—I cannot say which of these statements is best supported by actual observation; for none of the writers I have quoted gives any reference to the original memoir from which his statement is derived. As far as my own reading goes, I have only come across one original record of experiments, made directly on the flash of lightning itself, with a view to determine the period of its duration. These experiments were carried out by Professor Ogden Rood, of Columbia

College, New York, between the years 1870 and 1873, and are recorded in the *American Journal of Science and Arts*.^[6]

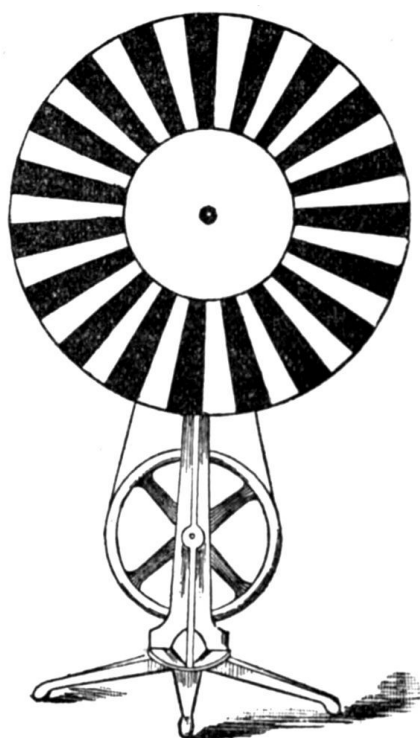
For the description of his apparatus, and for the details of his observations, I must refer you to the memoir itself; but I may tell you briefly that the results at which he arrived, if they be accepted, must lead to a considerable modification of the views previously entertained on the subject. In the first place, he satisfied himself that what appears to the eye a single flash of lightning is usually, if not always, multiple in its character; consisting, in fact, of a succession of distinct flashes, which follow one another with such rapidity as to make a continuous impression on the retina. Next, he proceeded to measure approximately the duration of these several component flashes; and he found that it varied over a wide range, amounting sometimes to fully the twentieth of a second, and being sometimes less than the sixteen-hundredth of a second.

Wheatstone's Experiments.—These results are extremely interesting; but we can hardly regard them as finally established, until they have been confirmed by other observers. I may remark, however, that they fit in very well with the experiments made by Professor Wheatstone, many years ago, on the duration of the electric spark, which, as I told you, is a miniature flash of lightning. In these classical experiments, which leave nothing to be desired in point of accuracy, Professor Wheatstone showed that a spark taken directly from a Leyden jar, or a spark taken from the conductor of a powerful electric machine, that is, just such a spark as you have seen here to-day, lasts for less than the millionth of a second.

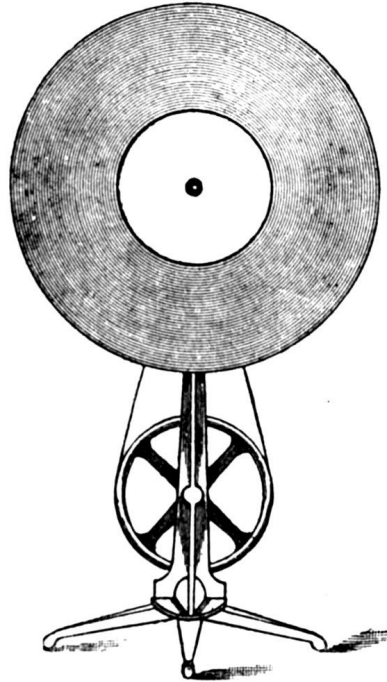
But he also showed that the duration of the spark is greatly increased, when a resisting wire is introduced into the path of the discharge. Thus, for example, when the discharge from a Leyden jar was made to pass through half a mile of copper wire, with breaks at intervals, the sparks that appeared at these breaks were found to last for $\frac{1}{24000}$ of a second.^[7] Hence we should naturally expect that the period of illumination would be still further increased, in the case of a flash of lightning, where the resistance interposed is enormously greater than in either of the experiments made by Wheatstone.^[8]

Experiment of the Rotating Disk.—It would be tedious, on an occasion like the present, to enter into an account of Wheatstone's beautiful and

ingenious method of investigation, by which the above facts have been established; but I will show you a much more simple experiment which brings home very forcibly to the mind how exceedingly short must be the duration of the electric spark. Here is a circular disk of cardboard, the outer part of which, as you see, is divided into sectors, black and white alternately, while the space about the centre is entirely white. The disk is mounted on a stand, by means of which I can make it rotate with great velocity. When it is put in rotation, the effect on the eye is very striking—the central space remains white as before, but in the outer rim the distinction of black and white absolutely disappears and gives place to a uniform gray. This color is due to the blending together of black and white in equal proportions; the blending being effected, not on the cardboard disk, but on the retina of the eye.



CARDBOARD DISK AS SEEN WHEN AT REST.



SAME DISK AS SEEN WHEN IN RAPID ROTATION.

I mentioned just now that an impression made on the retina lasts for the tenth of a second after the cause of it has been removed. Now, when this disk is in rotation, the sectors follow one another so rapidly that the particular part of space occupied at any moment by a white sector will be occupied by a black sector within a time much less than the tenth of a second. It follows that the impression made by each white sector remains on the retina until the following black sector comes into the same position; and, in like manner, the impression made by each black sector remains until the following white sector takes up the position of the black. Therefore, the impression made by the whole outer rim is the impression of black and white combined—that is, the impression of gray.

So far, I dare say, the phenomenon is already familiar to you all. But I propose now to show you the revolving disk illuminated by the electric spark; and you will observe that, at the moment of illumination, the black and white sectors come out as clearly and distinctly as if the disk were standing still.

For the success of this experiment it is desirable, not only to have a brilliant spark in order to secure a good illumination of the disk, but also to have a succession of such sparks, that you may see the phenomenon

frequently repeated, and thus be able to observe it at your leisure. To attain these two objects, I have made the arrangement which is here before you.

In front of the disk is a large and very powerful Leyden jar. The rod connected with the inner coating rises well above the mouth of the jar, and ends in a brass ball nearly opposite the centre of the disk. Connected with the outer coating of the jar is another rod which likewise ends in a brass ball, and which is so adjusted that the distance between the two balls is about an inch. The two rods are connected respectively with the two conductors of a Holtz machine, so that, when the machine is worked, the jar is first quickly charged, and then it discharges itself, with a brilliant spark, between the two brass balls. Thus, by continuing to work the machine, we can get, as long as we choose, a succession of sparks following one another at short and regular intervals right in front of the disk.

Everything being now ready, and the room partially darkened, the disk is put in rapid rotation; and you can see, by the twilight that remains, the outer rim a uniform gray, and the central space white. But when my assistant begins to turn the Holtz machine, and brilliant sparks leap out at intervals, the revolving disk, illuminated for a moment at each discharge, seems to be standing still, and shows the black and white sectors distinctly visible.

The reason of this is clear: So brief is the moment for which the spark endures, that the disk, though in rapid motion, makes no sensible advance during that small fraction of time; therefore, in the image on the retina, the impression made by the white sectors remains distinct from the impression made by the black, and the eye sees the disk as it really is.

I may notice, in passing, a very interesting consideration, suggested by this experiment. A cannon ball is now commonly discharged with a velocity of about 1,600 feet a second. Moving with this velocity it is, as you know, under ordinary circumstances, altogether invisible to the eye. But suppose it were illuminated, in the darkness of night, by this electric spark, which lasts, we will say, for the millionth of a second. During the moment of illumination, the cannon ball moves through the millionth part of 1,600 feet, which is a little less than the fiftieth of an inch. Practically, we may say that the cannon ball does not sensibly change its place while the spark lasts. Further, the impression it makes on the eye, from the position it occupies at the moment of illumination, remains on the retina for at least the tenth of a

second. Therefore, if we are looking toward that particular part of space where the cannon ball happens to be at the moment the spark passes, we must see the cannon ball hanging motionless in the air, though we know it is traveling at the rate of 1,600 feet a second, or about 1,000 miles an hour.

Brightness of a Flash of Lightning.—I should like to say one word about the brightness of a flash of lightning. Somewhat more than thirty years ago, Professor Swan, of Edinburgh, showed that the eye requires a sensible time—about the tenth of a second—to perceive the full brightness of a luminous object. Further, he proved, by a series of interesting experiments, that when a flash of light lasts for less than the tenth of a second, its apparent brilliancy to the eye is proportional to the time of its duration.^[9] Now consider the consequence of these facts in reference to the brightness of our electric spark. If the spark lasted for the tenth of a second, we should perceive its full brightness; if it lasted for the tenth part of that time, we should see only the tenth part of its brightness; if it lasted for the hundredth part, we should see only the hundredth part of its brightness; and so on. But we know, in point of fact, that it lasts for less than the millionth of a second, that is, less than the hundred-thousandth part of the tenth of a second. Therefore we see only the hundred-thousandth part of its real brightness.

Here is a startling conclusion, and one, I may say, fully justified by scientific evidence. That electric spark, brilliant as it appears to us, is really a hundred thousand times as bright as it seems to be. We cannot speak with the same precision of a flash of lightning; because its duration has not yet been so exactly determined. But if we suppose that a flash of lightning, in a particular case, lasts for the thousandth of a second, it would follow, from the above experiments, that the flash is a hundred times as bright, in fact, as it appears to the eye.

Various Forms of Lightning.—The lightning of which I have spoken hitherto is commonly called *forked* lightning; a name which seems to have been derived from the zig-zag line of light it presents to the eye. But there are other forms under which the electricity of the clouds often makes itself manifest; and to these I would now invite your attention for a few moments. The most common of them all, at least in this country, is that which is familiarly known by the name of *sheet* lightning. This is, probably, nothing

else than the lighting up of the atmosphere, or of the clouds, by forked lightning, which is not itself directly visible.

Generally speaking, after a flash of sheet lightning, we hear the rolling of distant thunder. But it sometimes happens, especially in summer time, that the atmosphere is again and again lit up by a sudden glow of light, and yet no thunder is heard. This phenomenon is commonly called *summer lightning*, or *heat lightning*. It is probably due, in many cases, to electrical discharges in the higher regions of the atmosphere, where the air is greatly rarified; and, in these cases, it would seem to resemble the discharges obtained by means of an induction coil in glass tubes containing rarified gases. But there is little doubt that in many cases, too, summer lightning, like ordinary sheet lightning, is due to forked lightning, which is so remote that we can neither see the flash itself directly, nor hear the rolling of the thunder.

Perhaps the most distinct and satisfactory evidence on this subject, derived from actual observation, is contained in the following letter of Professor Tyndall, written in May, 1883: “Looking to the south and south-east from the Bel Alp, the play of silent lightning among the clouds and mountains is sometimes very wonderful. It may be seen palpitating for hours, with a barely appreciable interval between the thrills. Most of those who see it regard it as lightning without thunder—Blitz ohne Donner, Wetterleuchten, I have heard it named by German visitors. The Monte Generoso, overlooking the Lake of Lugano, is about fifty miles from the Bel Alp, as the crow flies. The two points are connected by telegraph; and frequently when the Wetterleuchten, as seen from the Bel Alp, was in full play, I have telegraphed to the proprietor of the Monte Generoso Hotel and learned, in every instance, that our silent lightning co-existed in time with a thunderstorm more or less terrific in upper Italy.”^[10]

Another form of lightning, described by many writers, is called *globe lightning*. It is said to appear as a ball of fire, about the size of a child’s head, or even larger, which moves for a time slowly about, and then, after the lapse of several seconds, explodes with a terrific noise, sending forth flashes of fire in all directions, which burn whatever they strike. Many accounts are on record of such phenomena; but they are derived, for the most part, from the evidence of persons who were not specially competent to observe, and to describe with precision, the facts that fell under their

observation. Hence these accounts, while they are accepted by some, are rejected by others; and it seems to me, in the present state of the question, that the existence of globe lightning can hardly be regarded as a demonstrated fact. At all events, if phenomena of this kind have really occurred, I can only say that nothing we know about electricity, at present, will enable us to account for them.^[11]

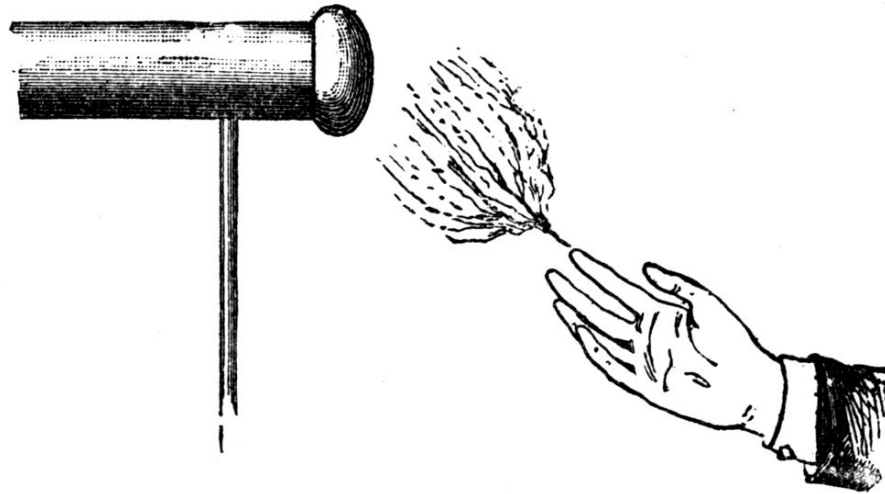
St. Elmo's Fire.—A much more authentic and, at the same time, very interesting form, under which the electricity of the clouds sometimes manifests its presence, is known by the name of St. Elmo's fire. This phenomenon at one time presents the appearance of a star, shining at the points of the lances or bayonets of a company of soldiers; at another, it takes the form of a tuft of bluish light, which seems to stream away from the masts and spars of a ship at sea, or from the pointed spire of a church. It was well known to the ancients. Cæsar, in his Commentaries, tells us that, after a stormy night, the iron points of the javelins of the fifth legion seemed to be on fire; and Pliny says that he saw lights, like stars, shining on the lances of the soldiers, keeping watch by night upon the ramparts. When two such lights appeared at once, on the masts of a ship, they were called Castor and Pollux, and were regarded by sailors as a sign of a prosperous voyage. When only one appeared, it was called Helen, and was taken as an unfavorable omen.

In modern times St. Elmo's fire has been witnessed by a host of observers, and all its various phases have been repeatedly described. In the memoirs of Forbin we read that, when he was sailing once, in 1696, among the Balearic Islands, a sudden storm came on during the night, accompanied by lightning and thunder. "We saw on the vessel," he says, "more than thirty St. Elmo's fires. Among the rest there was one on the vane of the mainmast more than a foot and a half high. I sent a man up to fetch it down. When he was aloft he cried out that it made a noise like wetted gunpowder set on fire. I told him to take off the vane and come down; but, scarcely had he removed it from its place, when the fire left it and reappeared at the end of the mast, so that it was impossible to take it away. It remained for a long time, and gradually went out."

On the 14th of January, 1824, Monsieur Maxadorf happened to look at a load of straw in the middle of a field just under a dense black cloud. The straw seemed literally on fire—a streak of light went forth from every

blade; even the driver's whip shone with a pale-blue flame. As the black cloud passed away, the light gradually disappeared, after having lasted about ten minutes. Again, it is related that on the 8th of May, 1831, in Algiers, as the French artillery officers were walking out after sunset without their caps, each one saw a tuft of blue light on his neighbor's head; and, when they stretched out their hands, a tuft of light was seen at the end of every finger. Not infrequently a traveler in the Alps sees the same luminous tuft on the point of his alpenstock. And quite recently, during a thunderstorm, a whole forest was observed to become luminous just before each flash of lightning, and to become dark again at the moment of the discharge. [\[12\]](#)

This phenomenon may be easily explained. It consists in a gradual and comparatively silent electrical discharge between the earth and the cloud; and generally, but not always, it has the effect of preventing such an accumulation of electricity as would be necessary to produce a flash of lightning. I can illustrate this kind of discharge with the aid of our machine. If I hold a pointed metal rod toward the large conductor, you can see, when the machine is worked and the room darkened, how the point of the rod becomes luminous and shines like a faint blue star. I substitute for the pointed rod the blunt handles of a pair of pliers, and a tuft of blue light is at once developed at the end of each handle, and seems to stream away with a hissing noise. I now put aside the pliers, and open out my hand under the conductor—and observe how I can set up, at pleasure, a luminous tuft at the tips of my fingers. Now and then a spark passes, giving me a smart shock, and showing how the electricity may sometimes accumulate so fast that it cannot be sufficiently discharged by the luminous tuft. Lastly, I present a small bushy branch of a tree to the conductor, and all its leaves and twigs are aglow with bluish light, which ceases for a moment when a spark escapes, to be again renewed when electricity is again developed by the working of the machine.



THE BRUSH DISCHARGE, ILLUSTRATING ST. ELMO'S FIRE.

Now, if you put a thundercloud in the place of that conductor, you can easily realize how, through its influence, the lance and bayonet of the soldier, the alpenstock of the traveler, the pointed spire of a church, the masts of a ship at sea, the trees of a forest, can all be made to glow with a silent electrical discharge which may or may not, according to circumstances, culminate at intervals in a genuine flash of lightning.

Origin of Lightning.—When we seek to account for the origin of lightning, we are confronted at once with two questions of great interest and importance—first, What are the sources from which the electricity of the thundercloud is derived? and, secondly, How does this electricity come to be developed in a form which so far transcends in power the electricity of our machines? These questions have long engaged the attention of scientific men, but I cannot say that they have yet received a perfectly satisfactory solution. Nevertheless, some facts of great scientific value have been established, and some speculations have been put forward, which are well deserving of consideration.

In the first place, it is quite certain that the atmosphere which surrounds our globe is almost always in a state of electrification. Further, the electrical condition of the atmosphere would seem to be as variable as the wind. It changes with the change of season; it changes from day to day; it changes from hour to hour. The charge of electricity is sometimes positive, sometimes negative; sometimes it is strong, sometimes feeble; and the

transition from one condition to another is sometimes slow and gradual, sometimes sudden and violent.

As a general rule, in fine, clear weather, the electricity of the atmosphere is positive, and not very strongly developed. In wet weather the charge may be either positive or negative, and is generally strong, especially when there are sudden heavy showers. In fog it is also strong, and almost always positive. In a snowstorm it is very strong, and most frequently positive. Finally, in a thunderstorm it is extremely strong, and generally negative; but it is subject to a sudden change of sign, when a flash of lightning passes or when rain begins to fall.

So far I have simply stated facts, which have been ascertained by careful observations, made at different stations by competent observers, and extending over a period of many years. But as regards the process by which the electricity of the atmosphere is developed, we have, up to the present time, no certain knowledge. It has been said that electricity may be generated in the atmosphere by the friction of the air itself, and of the minute particles floating in it, against the surface of the earth, against trees and buildings, against rocks, cliffs, and mountains. But this opinion, however probable it may be, has not yet been confirmed by any direct experimental investigation.

The second theory is that the electricity of the atmosphere is due, in great part at least, to the evaporation of salt water. Many years ago, Pouillet, a French philosopher, made a series of experiments in the laboratory, which seemed to show that evaporation is generally attended with the development of electricity; and, in particular, he satisfied himself that the vapor which passes off from the surface of salt water is always positively electrified. Now, the atmosphere is everywhere charged, more or less, with vapor which comes, almost entirely, from the salt water of the ocean. Hence Pouillet inferred that the chief source of atmospheric electricity is the evaporation of sea water. This explanation would certainly go far to account for the presence of electricity in the atmosphere, if the fact on which it rests were established beyond dispute. But there is some reason to doubt whether the development of electricity, in the experiments of Pouillet, was due simply to the process of evaporation, and not rather to other causes, the influence of which he did not sufficiently take into account.

A conjecture has recently been started that electricity may be generated by the mere impact of minute particles of water vapor against minute particles of air.^[13] If this conjecture could be established as a fact, it would be amply sufficient to account for all the electricity of the atmosphere. From the very nature of a gas, the molecules of which it is composed are forever flying about with incredible velocity; and therefore the particles of water vapor and the particles of air, which exist together in the atmosphere, must be incessantly coming into collision. Hence, however small may be the charge of electricity developed at each individual impact, the total amount generated over any considerable area, in a single day, must be very great indeed. It is evident, however, that this method of explaining the origin of atmospheric electricity can only be regarded as, at best, a probable hypothesis, until the assumption on which it rests is supported by the evidence of observation or experiment.

Length of a Flash of Lightning.—It would seem, then, that we are not yet in a position to indicate with certainty the sources from which the electricity of the atmosphere is derived. But whatever these sources may be, there can be little doubt that the electricity of the atmosphere is intimately associated with the minute particles of water vapor of which the thundercloud is eventually built up. This consideration is of great importance when we come to consider the special properties of lightning, as compared with other forms of electricity. The most striking characteristic of lightning is the wonderful power it possesses of forcing its way through the resisting medium of the air. In this respect it incomparably surpasses all forms of electricity that have hitherto been produced by artificial means. The spark of an ordinary electric machine can leap across a space of three or four inches; the machine we have employed in our experiments to-day can give, under favorable circumstances, a spark of nine or ten inches; the longest electric spark ever yet produced artificially is probably the spark of Mr. Spottiswoode's gigantic induction coil; and it does not exceed three feet six inches. But the length of a flash of lightning is not to be measured in inches, or in feet or in yards; it varies from one or two miles, for ordinary flashes, to eight or ten miles in exceptional cases.

This power of discharging itself violently through a resisting medium, in which the thundercloud so far transcends the conductor of an electric machine, is due to the property commonly known among scientific men as

electrical *potential*. The greater the distance to which an electrified body can shoot its flashes through the air, the higher must be its potential. Hence the potential of a thundercloud must be exceedingly high, since its flashes can pierce the air to a distance of several miles. And what I want to point out is, that we are able to account for this exceedingly high potential, if we may only assume that the minute particles of water vapor in the atmosphere have, from any cause, received ever so small a charge of electricity. The number of such particles that go to make up an ordinary drop of rain are to be counted by millions of millions; and it is capable of scientific proof that, as each new particle is added, in the building up of the drop, a rise of potential is necessarily produced. It is clear, therefore, that there is practically no limit to the potential that may be developed by the simple agglomeration of very small cloud particles, each carrying a very small charge of electricity.^[14]

This explanation, which traces the exceedingly high potential of lightning to the building up of rain drops in the thundercloud, suggests a reason why it so often happens that immediately after a flash of lightning “the big rain comes dancing to the earth.” The potential has been steadily rising as the drops have been getting larger and larger, until at length the potential has become so high that the thundercloud is able to discharge itself, and almost at the same moment the drops have become so large that they can no longer be held aloft against the attracting force of gravity.

Physical Cause of Thunder.—Let us now proceed to consider the phenomenon of thunder, which is so intimately associated with lightning, and which, though perfectly harmless in itself, and though never heard until the real danger is past, often excites more terror in the mind than the lightning flash itself. The sound of thunder, like that of the electric spark, is due to a disturbance caused in the air by the electric discharge. The air is first expanded by the intense heat that is developed along the line of discharge, and then it rushes back again to fill up the partial vacuum which its expansion has produced. This sudden movement gives rise to a series of sound waves, which reach the ear in the form of thunder. But there are certain peculiar characteristics of thunder which are deserving of special consideration.

Rolling of Thunder.—They may be classified, I think, under two heads. First, the sound of thunder is not an instantaneous report like the sound of

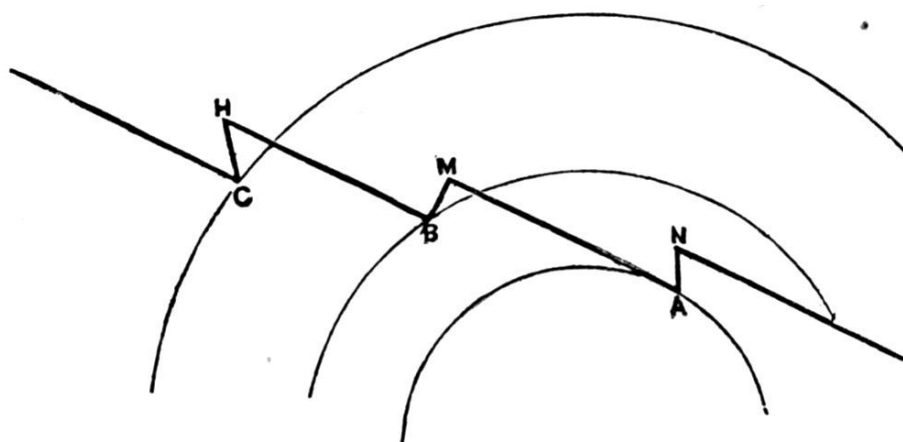
the electric spark—it is a prolonged peal lasting, sometimes, for several seconds. Secondly, each flash of lightning gives rise, not to one peal only, but to a succession of peals following one another at irregular intervals. These two phenomena, taken together, produce that peculiar effect on the ear which is commonly described as the *rolling* of thunder; and both of them, I think, may be sufficiently accounted for in accordance with the well-established properties of sound.

To understand why the sound of thunder reaches the ear as a prolonged peal, we have only to remember that sound takes time to travel. Since a flash of lightning is practically instantaneous, we may assume that the sound is produced at the same moment all along the line of discharge. But the sound waves, setting out at the same moment from all points along the line of discharge, must reach the ear in successive instants of time, arriving first from that point which is nearest to the observer, and last from that point which is most distant. Suppose, for example, that the nearest point of the flash is a mile distant from the observer, and the farthest point two miles—the sound will take about five seconds to come from the nearest point, and about ten seconds to come from the farthest point; and moreover, in each successive instant from the time the first sound reaches the ear, sound will continue to arrive from the successive points between. Therefore the thunder, though instantaneous in its origin, will reach the ear as a prolonged peal extending over a period of five seconds.

Succession of Peals.—The succession of peals produced by a single flash of lightning is due to several causes, each one of which may contribute more or less, according to circumstances, toward the general effect. First, if we accept the results arrived at by Professor Ogden Rood, of Columbia College, what appears to the eye as a single flash of lightning, consists, in fact, as a general rule, of a succession of flashes, each one of which must naturally produce its own peal of thunder; and although the several flashes, if they follow one another at intervals of the tenth of a second, will make one continuous impression on the eye, the several peals of thunder, under the same conditions, will impress the ear as so many distinct peals.

The next cause that I would mention is the zigzag path of the lightning discharge. To make clear to you the influence of this circumstance, I must ask your attention for a moment to the [diagram](#) on next page. Let the broken

line represent the path of a flash of lightning, and let o represent the position of an observer. The sound will reach him first from the point A , which is nearest to him, and then it will continue to arrive in successive instants from the successive points along the line $A N$ and along the line $A M$, thus producing the effect of a continuous peal. Meanwhile the sound waves have been traveling from the point B , and in due time will reach the observer at o . Coming as they do in a different direction from the former, they will strike the ear as the beginning of a new peal which, in its turn, will be prolonged by the sound waves arriving, in successive instants, from the successive points along the line $B M$ and $B H$. A little later, the sound will arrive from the more distant point C , and a third peal will begin. And so there will be several distinct peals proceeding, so to speak, from several distinct points in the path of the lightning flash.



ORIGIN OF SUCCESSIVE PEALS OF THUNDER.

A third cause to which the succession of peals may be referred is to be found in the minor electrical discharges that must often take place within the thundercloud itself. A thundercloud is not a continuous mass like the metal cylinder of this electric machine—it has many outlying fragments, more or less imperfectly connected with the principal body. Moreover, the material of which the cloud is composed is only a very imperfect conductor as compared with our brass cylinder. For these two reasons it must often happen, about the time a flash of lightning passes, that different parts of the cloud will be in such different electrical conditions as to give rise to electrical discharges within the cloud itself. Each of these discharges

produces its own peal of thunder; and thus we may have a number of minor peals, sometimes preceding and sometimes following the great crash which is due to the principal discharge.

Lastly, the influence of echo has often a considerable share in multiplying the number of peals of thunder. The waves of sound, going forth in all directions, are reflected from the surfaces of mountains, forests, clouds, and buildings, and coming back from different quarters, and with varying intensity, reach the ear like the roar of distant artillery. The striking effect of these reverberations in a mountain district has been described by a great poet in words which, I daresay, are familiar to most of you:

“Far along,
From peak to peak, the rattling crags among,
Leaps the live thunder! Not from one lone cloud,
But every mountain now has found a tongue,
And Jura answers from her misty shroud
Back to the joyous Alps, that call to her aloud!”

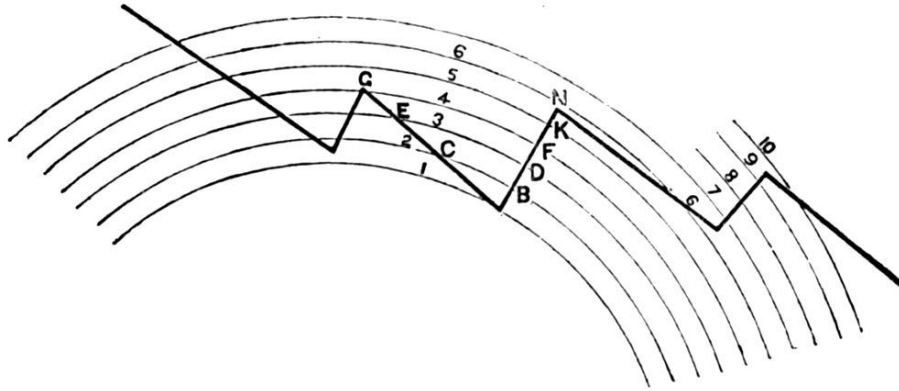
Variations of Intensity in Thunder.—From what has been said, it is easy to understand how the general roar of thunder is subject to great changes of intensity, during the time it lasts, according to the number of peals that may be arriving at the ear of an observer in each particular moment. But every one must have observed that even an individual peal of thunder often undergoes similar changes, swelling out at one moment with great power, and the next moment rapidly dying away. To account for this phenomenon, I would observe, first, that there is no reason to suppose that the disturbance caused by lightning is of exactly the same magnitude at every point of its path. On the contrary, it would seem very probable that the amount of this disturbance is, in some way, dependent on the resistance which the discharge encounters. Hence the intensity of the sound waves sent forth by a flash of lightning is probably very different at different parts of its course; and each individual peal will swell out on the ear or die away, according to the greater or less intensity of the sound waves that reach the ear in each successive moment of time.

But there is another influence at work which must produce variations in the loudness of a peal of thunder, even though the sound waves, set in motion by the lightning, were everywhere of equal intensity. This influence

depends on the position of the observer in relation to the path of the lightning flash. At one part of its course the lightning may follow a path which remains for a certain length at nearly the same distance from the observer; then all the sound produced along this length will reach the observer nearly at the same moment, and will burst upon the ear with great intensity. At another part, the lightning may for an equal length go right away from the observer; and it is evident that the sound produced along this length will reach the observer in successive instants, and consequently produce an effect comparatively feeble.

With a view to investigate this interesting question a little more closely, let me suppose the position of the observer taken as a centre, and a number of concentric circles drawn, cutting the path of the lightning flash, and separated from one another by a distance of 110 feet, measured along the direction of the radius. It is evident that all the sound produced between any two consecutive circles will reach the ear within a period which must be measured by the time that sound takes to travel 110 feet, that is, within the tenth of a second. Hence, in order to determine the quantity of sound that reaches the ear in successive periods of one-tenth of a second, we have only to observe how much is produced between each two consecutive circles. But on the supposition that the sound waves, set in motion by the flash of lightning, are of equal intensity at every point of its path, it is clear that the quantity of sound developed between each two consecutive circles will be simply proportional to the length of the path enclosed between them.

With these principles established, let us now follow the course of a peal of thunder, in the diagram before us. This broken line, drawn almost at random, represents the path of a flash of lightning; the observer is supposed to be placed at o, which is the centre of the concentric circles; these circles are separated from one another by a distance of 110 feet, measured in the direction of the radius; and we want to consider how any one peal of thunder may vary in loudness in the successive periods of one-tenth of a second.



VARIATIONS OF INTENSITY IN A PEAL OF THUNDER.

Let us take, for example, the peal which begins when the sound waves reach the ear from the point A. In the first unit of time the sound that reaches the ear is the sound produced along the lines A B and A C; in the second unit, the sound produced along the lines B D and C E; in the third unit, the sound produced along D F and E G. So far the peal has been fairly uniform in its intensity; though there has been a slight falling off in the second and third units of time, as compared with the first. But in the fourth unit there is a considerable falling away of the sound; for the line F K is only about one-third as long as D F and E G taken together; therefore the quantity of sound that reaches the ear in the fourth unit of time is only one-third of that which reaches it in each of the three preceding units; and consequently the sound is only one-third as loud. In the fifth unit, however, the peal must rise to a sudden crash; for the portion of the lightning path inclosed between the fifth and sixth circles is about six times as great as that between the fourth and fifth; therefore the intensity of the sound will be suddenly increased about six-fold. After this sudden crash, the sound as suddenly dies away in the sixth unit of time; it continues feeble as the path of the lightning goes nearly straight away from the observer; it swells again slightly in the ninth unit of time; and then continues without much variation to the end. This is only a single illustration, but it seems quite sufficient to show that the changes of intensity in a peal of thunder must be largely due to the position of the spectator in relation to the several parts of the lightning flash.

Distance of a Flash of Lightning.—I need hardly remind you that, by observing the interval that elapses between the flash of lightning and the peal of thunder that follows it, we may estimate approximately the distance of the nearest point of the discharge. Light travels with such amazing

velocity that we may assume, without any sensible error, that we see the flash of lightning at the very moment in which the discharge takes place. But sound, as we have seen, takes a sensible time to travel even short distances; and therefore a measurable interval almost always elapses between the moment in which the flash is seen and the moment in which the peal of thunder first reaches the ear. And the distance through which sound travels in this interval will be the distance of the nearest point through which the discharge has passed. Now, the velocity of sound in air varies slightly with the temperature; but, at the ordinary temperature of our climate, we shall not be far astray if we allow 1,100 feet for every second, or about one mile for every five seconds.

You will observe also that, by repeating this observation, we can determine whether the thundercloud is coming toward us, or going away from us. So long as the interval between each successive flash and the corresponding peal of thunder, continues to get shorter and shorter, the thundercloud is approaching; when the interval begins to increase, the thundercloud is receding from us, and the danger is passed.

The crash of thunder is terrific when the lightning is close at hand; but it is a curious fact, that the sound does not seem to travel as far as the report of an ordinary cannon. We have no authentic record of thunder having been heard at a greater distance than from twelve to fifteen miles, whereas the report of a single cannon has been heard at five times that distance; and the roar of artillery, in battle, at a greater distance still. On the occasion of the Queen's visit to Cherbourg, in August, 1858, the salute fired in honor of her arrival was heard at Bonchurch, in the Isle of Wight, a distance of sixty miles. It was also heard at Lyme Regis, in Dorsetshire, which is eighty-five miles from Cherbourg, as the crow flies; and we are told that, not only was it audible in its general effect, but the report of individual guns was distinctly recognized. The artillery of Waterloo is said to have been heard at the town of Creil, in France, 115 miles from the field of battle; and the cannonading at the siege of Valenciennes, in 1793, was heard, from day to day, at Deal, on the coast of England, a distance of 120 miles. ^[15]

So far, I have endeavored to set forth some general ideas on the nature and origin of lightning, and of the thunder that accompanies it. In my next Lecture I propose to give a short account of the destructive effects of

lightning, and to consider how these effects may best be averted by means of lightning conductors.

NOTE TO PAGE 20.

ON THE HIGH POTENTIAL OF A FLASH OF LIGHTNING.

The potential of an electrified sphere is equal to the quantity of electricity with which the sphere is charged, divided by the radius of the sphere. Now the minute cloud particles, which go to make up a drop of rain, may be taken to be very small spheres; and if v represent the potential of each one, q the quantity of electricity with which it is charged, and r the radius of the sphere, we have $v = q/r$. Suppose 1,000 of these cloud particles to unite into one; the quantity of electricity in the drop, thus formed, will be $1,000q$; and the radius, which increases in the ratio of the cube root of the volume, will be $10r$. Therefore the potential of the new sphere will be $1000q/10r$, or $100q/r$; that is to say, it will be 100 times as great as the potential of each of the cloud particles which compose it. When a million of cloud particles are blended into a single drop, the same process will show that the potential has been increased ten thousandfold; and when a drop is produced by the agglomeration of a million of millions of cloud particles, the potential of the drop will be a hundred million times as great as that of the individual particles. ^[16]

FOOTNOTES:

[1] “Il y a des grands seigneurs dont il ne faut approcher qu’avec d’extrêmes précautions. Le tonnerre est de ce nombre.”—Dict. Philos. art. Foudre.

[2] *Electricité Statique*, ii., 561.

[3] *Deschanel’s Natural Philosophy*, Sixth Edition, p. 641.

[4] *Fragments of Science*, Fifth Edition, p. 311.

[5] *Lecture on Thunderstorms*, *Nature*, vol. xxii., p. 341.

[6] *Third Series*, vol. v., p. 161.

[7] Phil. Trans. Royal Society, 1834, vol. cxxv., pp. 583-591.

[8] In experiments with a Leyden jar, Feddersen has shown that the duration of the discharge is increased, not only by increasing the striking distance, but also by increasing the size of the jar. Now, a flash of lightning may be regarded as the discharge of a Leyden jar of immense size, with an enormous striking distance; and therefore we should expect that the duration of the discharge should be greatly prolonged. See *American Journal of Science and Arts*, Third Series, vol. i., p. 15.

[9] See original paper by Swan, Trans. Royal Society, Edinburgh, 1849, vol. xvi., pp. 581-603; also, a second paper, *ib.* 1861, vol. xxii., pp. 33-39.

[10] *Nature*, vol. xxviii., p. 54.

[11] See, however, an attempt to account for this phenomenon in De Larive's *Treatise on Electricity*, London, 1853-8, vol. iii., pp. 199, 200; and another, quite recently, by Mr. Spottiswoode, in a *Lecture on the Electrical Discharge*, delivered before the British Association at York, in September, 1881, and published by Longmans, London, p. 42. See also, for recent evidence regarding the phenomenon itself, Scott's *Elementary Meteorology*, pp. 175-8.

[12] See Jamin, "Cours de Physique," i., 480-1; Tomlinson, "The Thunderstorm," Third Edition, pp. 95-103; "Thunderstorms," a Lecture by Professor Tait, *Nature*, vol. xxii., p. 356.

[13] Professor Tait, *On Thunderstorms*, *Nature*, vol. xxii., pp. 436-7.

[14] See note at the end of this Lecture, p. 26.

[15] See Tomlinson, *The Thunderstorm*, pp. 87-9.

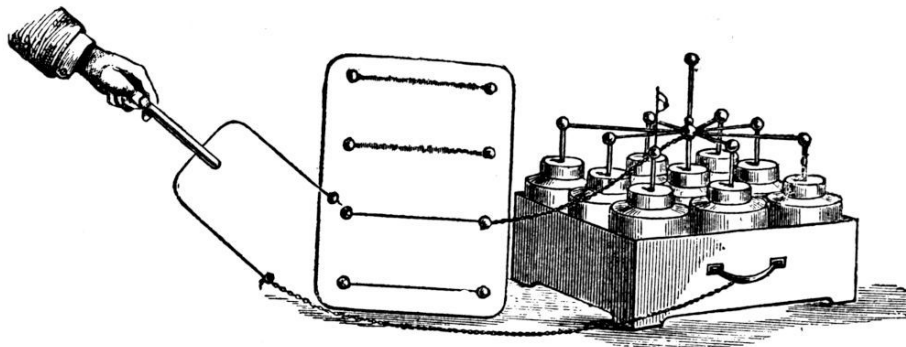
[16] See Tait on Thunderstorms, *Nature*, vol. xxii., p. 436.

LECTURE II.

LIGHTNING CONDUCTORS.

The effects of lightning, on the bodies that it strikes, are analogous to those which may be produced by the discharge of our electric machines and Leyden jar batteries. When the discharge of a battery traverses a metal conductor of sufficient dimensions to allow it an easy passage, it makes its way along silently and harmlessly. But if the conductor be so thin as to offer considerable resistance, then the conductor itself is raised to intense heat, and may be melted, or even converted into vapor, by the discharge.

On opposite page is shown a board on which a number of very thin wires have been stretched, over white paper, between brass balls. The wires are so thin that the full charge of the battery before you, which consists of nine large Leyden jars, is quite sufficient to convert them in an instant into vapor. I have already, on former occasions, sent the charge through two of these wires, and nothing remains of them now but the traces of their vapor, which mark the path of the electric discharge from ball to ball. At the present moment the battery stands ready charged, and I am going to discharge it through a third wire, by means of this insulated rod which I hold in my hand. The discharge has passed; you saw a flash, and a little smoke; and now, if you look at the paper, you will find that the wire is gone, but that it has left behind the track of its incandescent vapor, marking the path of the discharge.



DISCHARGE OF LEYDEN JAR BATTERY THROUGH THIN WIRES.

Destruction of Buildings by Lightning.—We learn from this experiment that the electricity stored up in our battery passes, without visible effect, through the stout wire of a discharging rod, but that it instantly converts into vapor the thin wire stretched across the spark board. And so it is with a flash of lightning. It passes harmlessly, as every one knows, through a stout metal rod, but when it comes across bell wires or telegraph wires, it melts them, or converts them into vapor. On the sixteenth of July, 1759, a flash of lightning struck a house in Southwark, on the south side of London, and followed the line of the bell wire. After the lightning had passed, the wire was no longer to be found; but the path of the lightning was clearly marked by patches of vapor which were left, here and there, adhering to the surface of the wall. In the year 1754, the lightning fell on a bell tower at Newbury, in the United States of America, and having dashed the roof to pieces, and scattered the fragments about, it reached the bell. From this point it followed an iron wire, about as thick as a knitting needle, melting it as it passed along, leaving behind a black streak of vapor on the surface of the walls.

Again, the electric discharge, passing through a bad conductor, produces mechanical disturbance, and, if the substance be combustible, often sets it on fire. So, too, as you know, the lightning flash, falling on a church spire, dashes it to pieces, knocking the stones about in all directions, while it sets fire to ships and wooden buildings; and more than once it has caused great devastation by exploding powder magazines.

Let me give you one or two examples: In January, 1762, the lightning fell on a church tower in Cornwall, and a stone—three hundred-weight—was torn from its place and hurled to a distance of 180 feet, while a smaller stone was projected as far as 1,200 feet from the building. Again, in 1809, the lightning struck a house not far from Manchester, and literally moved a massive wall twelve feet high and three thick to a distance of several feet. You may form some conception of the enormous force here brought into action, when I tell you that the total weight of mason-work moved on this occasion was not less than twenty-three tons.

The church of St. George, at Leicester, was severely damaged by lightning on the 1st of August, 1846. About 8 o'clock in the evening the rector of the parish saw a vivid streak of light darting with incredible velocity against the upper part of the spire. "For the distance of forty feet on the eastern side, and nearly seventy on the west, the massive stonework of the spire was instantly rent asunder and laid in ruins. Large blocks of stone were hurled in all directions, broken into small fragments, and in some cases, there is reason to believe, reduced to powder. One fragment of considerable size was hurled against the window of a house three hundred feet distant, shattering to pieces the woodwork, and strewing the room within with fine dust and fragments of glass. It has been computed that a hundred tons of stone were, on this occasion, blown to a distance of thirty feet in three seconds. In addition to the shivering of the spire, the pinnacles at the angles of the tower were all more or less damaged, the flying buttresses cracked through and violently shaken, many of the open battlements at the base of the spire knocked away, the roof of the church completely riddled, the roofs of the side entrances destroyed, and the stone staircases of the gallery shattered."^[17]

Lightning has been at all times the cause of great damage to property by its power of setting fire to whatever is combustible. Fuller says, in his Church History, that "scarcely a great abbey exists in England which once, at least, has not been burned by lightning from heaven." He mentions, as examples, the Abbey of Croyland twice burned, the Monastery of Canterbury twice, the Abbey of Peterborough twice; also the Abbey of St. Mary's, in Yorkshire, the Abbey of Norwich, and several others. Sir William Snow Harris, writing about twenty years ago, tells us that "the number of churches and church spires wholly or partially destroyed by lightning is beyond all belief, and would be too tedious a detail to enter upon. Within a comparatively few years, in 1822 for instance, we find the magnificent Cathedral of Rouen burned, and, so lately as 1850, the beautiful Cathedral of Saragossa, in Spain, struck by lightning during divine service and set on fire. In March of last year a dispatch from our Minister at Brussels, Lord Howard de Walden, dated the 24th of February, was forwarded by Lord Russell to the Royal Society, stating that, on the preceding Sunday, a violent thunderstorm had spread over Belgium; that twelve churches had been

struck by lightning; and that three of these fine old buildings had been totally destroyed.”^[18]

Even in our own day the destruction caused by fires produced through the agency of lightning is very great—far greater than is commonly supposed. No general record of such fires is kept, and consequently our information on the subject is very incomplete and inexact. I may tell you, however, one small fact which, so far as it goes, is precise enough and very significant. In the little province of Schleswig-Holstein, which occupies an area less than one-fourth of the area of Ireland, the Provincial Fire Assurance Association has paid in sixteen years, for damage caused by lightning, somewhat over £100,000, or at the rate of more than £6,000 a year. The total loss of property every year in this province, due to fires caused by lightning, is estimated at not less than £12,500.^[19]

Destruction of Ships at Sea.—The destructive effects of lightning on ships at sea, before the general adoption of lightning conductors, seems almost incredible at the present day. From official records it appears that the damage done to the Royal Navy of England alone involved an expenditure of from £6,000 to £10,000 a year. We are told by Sir William Snow Harris, who devoted himself for many years to this subject with extraordinary zeal and complete success, that between the year 1810 and the year 1815—that is, within a period of five years—“no less than forty sail of the line, twenty frigates, and twelve sloops and corvettes were placed *hors de combat* by lightning. In the merchant navy, within a comparatively small number of years, no less than thirty-four ships, most of them large vessels with rich cargoes, have been totally destroyed—been either burned or sunk—to say nothing of a host of vessels partially destroyed or severely damaged.”^[20]

And these statements, be it observed, take no account of ships that were simply reported as missing, some of which, we can hardly doubt, were struck by lightning in the open sea, and went down with all hands on board. A famous ship of forty-four guns, the *Resistance*, was struck by lightning in the Straits of Malacca, and the powder magazine exploding, she went to the bottom. Of her whole crew only three were saved, who happened to be picked up by a passing boat. It has been well observed that, were it not for these three chance survivors, nothing would have been known concerning

the fate of the vessel, and she would have been simply recorded as missing in the Admiralty lists.

Nothing is more fearful to contemplate than the scene on board a ship when she is struck by lightning in the open sea, with the winds howling around, the waves rolling mountains high, the rain coming down in torrents, and the vivid flashes lighting up the gloom at intervals, and carrying death and destruction in their track. I will read you one or two brief accounts of such a scene, given in the pithy but expressive language of the sailor. In January, 1786, the *Thisbe*, of thirty-six guns, was struck by lightning off the coast of Scilly, and reduced to the condition of a wreck. Here is an extract from the ship's log: "Four A. M., strong gales; handed mainsail and main top-sail; hove to with storm staysails; blowing very heavy, S. E. 4.15, a flash of lightning, with tremendous thunder, disabled some of our people. A second flash set the mainsail, main-top, and mizen staysails on fire. Obligated to cut away the mainmast; this carried away mizen top-mast and fore top-sail yard. Found foremast also shivered by the lightning. Fore top-mast went over the side about 9 A. M. Set the foresail."[\[21\]](#)

A few years later, in March, 1796, the *Lowestoffe* was struck in the Mediterranean, and we read as follows in the log of the ship: "North end of Minorca; heavy squalls; hail, rain, thunder, and lightning. 12.15, ship struck by lightning, which knocked three men from the masthead, one killed. 12.30, ship again struck; main top-mast shivered in pieces; many men struck senseless on the decks. Ship again struck, and set on fire in the masts and rigging; mainmast shivered in pieces; fore top-mast shivered; men benumbed on the decks, and knocked out of the top; one man killed on the spot. 1.30, cut away the mainmast; employed clearing wreck. 4, moderate; set the foresail."[\[22\]](#)

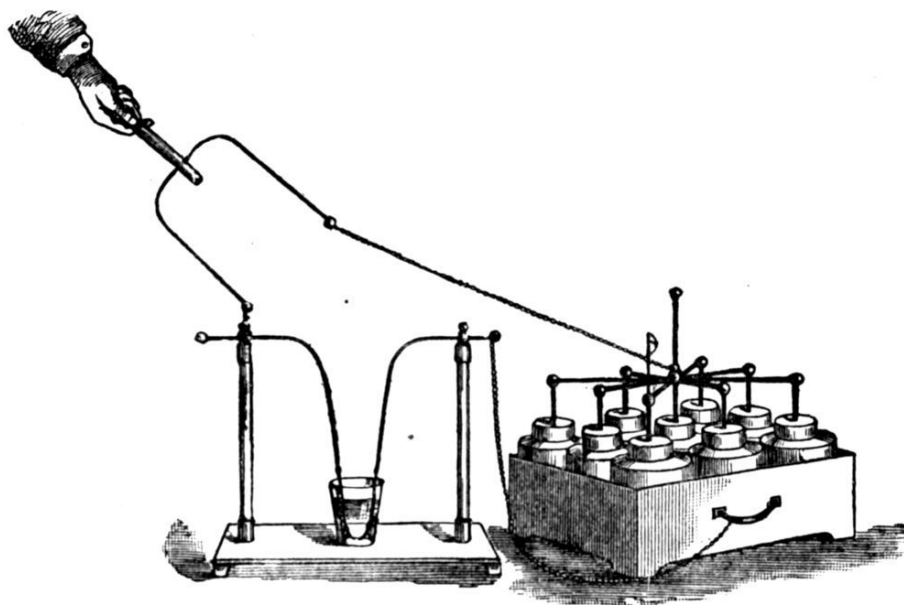
Again, in 1810, the *Repulse*, a ship of seventy-four guns, was struck, off the coast of Spain. "The wind had been variable in the morning—and at 12.35 there was a heavy squall, with rain, thunder, and lightning. The ship was struck by two vivid flashes of lightning, which shivered the maintop-gallant mast, and severely damaged the mainmast. Seven men were killed on the spot; three others only survived a few days; and ten others were maimed for life. After the second discharge the rain fell in torrents. The ship was more completely crippled than if she had been in action, and the

squadron, then engaged on a critical service, lost for a time one of its fastest and best ships.”^[23]

Destruction of Powder Magazines.—Not less appalling is the devastation caused by lightning when it falls on a powder magazine. Here is a striking example: On the eighteenth of August, 1769, the tower of St. Nazaire, at Brescia, was struck by lightning. Underneath the tower about 200,000 pounds of gunpowder, belonging to the Republic of Venice, were stored in vaults. The powder exploded, leveling to the ground a great part of the beautiful city of Brescia, and burying thousands of its inhabitants in the ruins. It is said that the tower itself was blown up bodily to a great height in the air, and came down in a shower of stones. This is, perhaps, the most fearful disaster of the kind on record. But we are not without examples in our own times. In the year 1856 the lightning fell on the Church of St. John, in the Island of Rhodes. A large quantity of gunpowder had been deposited in the vaults of the church. This was ignited by the flash; the building was reduced to a mass of ruins, a large portion of the town was destroyed, and a considerable number of the inhabitants were killed. Again, in the following year, the magazine of Joudpore, in the Bombay Presidency, was struck by lightning. Many thousand pounds of gunpowder were blown up, five hundred houses were destroyed, and nearly a thousand people are said to have been killed.^[24]

Experimental Illustrations.—And now, before proceeding further, I will make one or two experiments, with a view of showing that the electricity of our machines is capable of producing effects similar to those produced by lightning, though immeasurably inferior in point of magnitude. Here is a common tumbler, about three-quarters full of water. Into it I introduce two bent rods of brass, which are carefully insulated below the surface of the water by a covering of india-rubber. The points, however, are exposed, and come to within an inch of one another, near the bottom of the tumbler. Outside the tumbler, the brass rods are mounted on a stand, by means of which I can send the full charge of this Leyden jar battery through the water, from point to point. Since water is a bad conductor of electricity, as compared with metals, the charge encounters great resistance in passing through it, and in overcoming this resistance produces considerable mechanical commotion, which is usually sufficient to shiver the glass to pieces.

To charge the battery will take about twenty turns of this large Holtz machine. Observe how the pith ball of the electroscope rises as the machine is worked, showing that the charge is going in. And now it remains stationary; which is a sign that the battery is fully charged, and can receive no more. You will notice that the outside coating of the battery has been already connected with one of the brass rods dipping into the tumbler of water. By means of this discharger I will now bring the inside coating into connection with the other rod. And see, before contact is actually made, the spark has leaped across, and our tumbler is violently burst asunder from top to bottom.

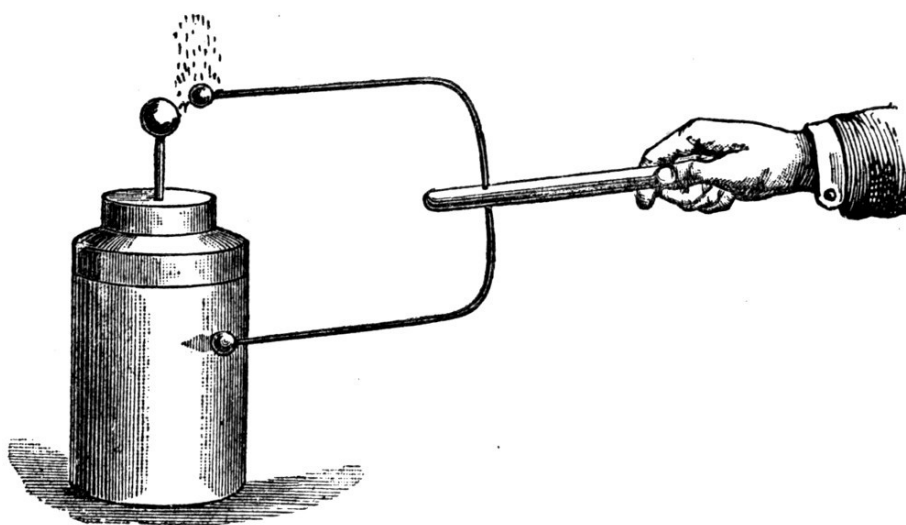


GLASS VESSEL BROKEN BY DISCHARGE OF LEYDEN JAR BATTERY.

This will probably appear to you a very small affair, when compared with the tearing asunder of solid masonry, and the hurling about of stones by the ton weight. No doubt it is; and that is just one of the lessons we have to learn from the experiment we have made. For, not only does it show us that effects of this kind may be caused by electricity artificially produced, but it brings home forcibly to the mind how incomparably more powerful is the lightning of the clouds than the electricity of our machines.

The property which electricity has of setting fire to combustible substances may be easily illustrated. This india-rubber tube is connected with the gas pipe under the floor, and to the end of the tube is fitted a brass stop-cock which I hold in my hand. I open the cock, and allow the jet of gas to flow toward the conductor of Carré's machine, while my assistant turns the handle; a spark passes, and the gas is lit. Again, my assistant stands on this insulating stool, placing his hand on the large conductor of the machine, while I turn the handle. His body becomes electrified, and when he presents his knuckle to this vessel of spirits of wine, which is electrically connected with the earth, a spark leaps across, and the spirits of wine are at once in a blaze. Once more; I tie a little gun-cotton around one knob of the discharging rod, and then use it to discharge a small Leyden jar; at the moment of the discharge the gun-cotton is set on fire.

It would be easy to explode gunpowder with the electric spark, but the smoke of the explosion would make the lecture-hall very unpleasant for the remainder of the lecture. I propose, therefore, to substitute for gunpowder an explosive mixture of oxygen and hydrogen, with which I have filled this little metal flask, commonly known as Volta's pistol. By a very simple contrivance, the electric spark is discharged through the mixture, when I hold the flask toward the conductor of the machine. A cork is fitted tightly into the neck of the flask, and at the moment the spark passes you hear a loud explosion, and you see the cork driven violently up to the ceiling.



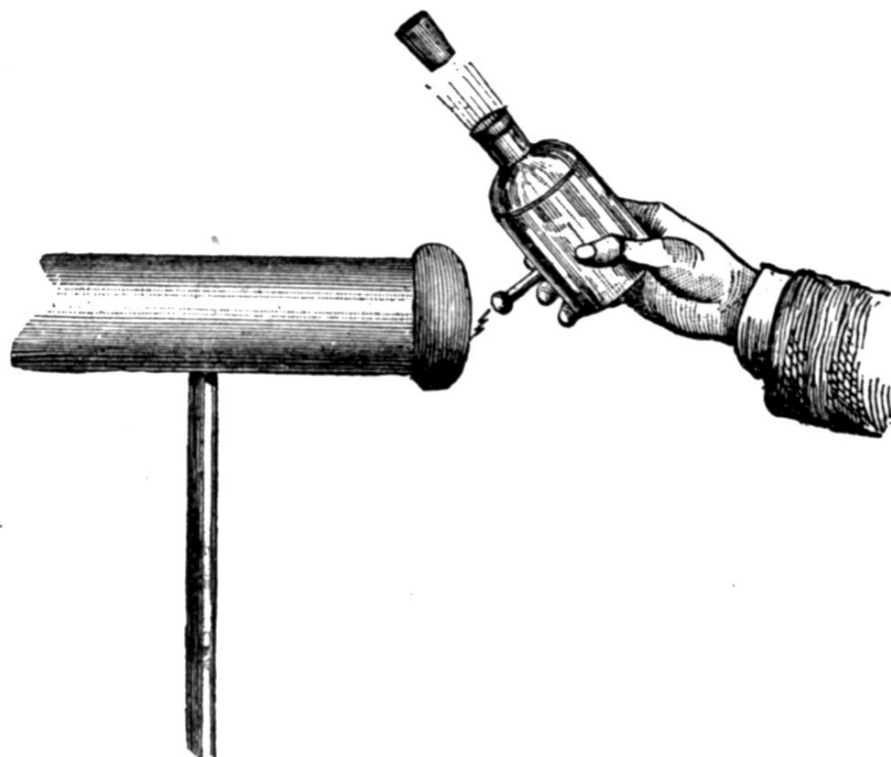
GUN-COTTON SET ON FIRE BY ELECTRIC SPARK.

Destruction of Life.—The last effect of lightning to which I shall refer, and which, perhaps, more than any other, strikes us with terror, is the sudden and utter extinction of life, when the lightning flash descends on man or on beast. So swift is this effect, in most cases, that death is, in all probability, absolutely painless, and the victim is dead before he can feel that he is struck. I cannot give you, with any degree of exactness, the number of people killed every year by lightning, because the record of such deaths has been hitherto very imperfectly kept, in almost all countries, and is, beyond doubt, very incomplete. But perhaps you will be surprised to learn that the number of deaths by lightning actually recorded is, on an average, in England about 22 every year, in France 80, in Prussia 110, in Austria 212, in European Russia 440.^[25]

So far as can be gathered from the existing sources of information, it would seem that the number of persons killed by lightning is, on the whole, about one in three of those who are struck. The rest are sometimes only stunned, sometimes more or less burned, sometimes made deaf for a time, sometimes partially paralyzed. On particular occasions, however, especially when the lightning falls on a large assembly of people, the number of persons struck down and slightly injured, in proportion to the number killed, is very much increased.

An interesting case of this kind is reported by Mr. Tomlinson. “On the twenty-ninth of August, 1847, at the parish church of Welton, Lincolnshire, while the congregation were engaged in singing the hymn before the sermon, and the Rev. Mr. Williamson had just ascended the pulpit, the lightning was seen to enter the church from the belfry, and instantly an explosion occurred in the centre of the edifice. All that could move made for the door, and Mr. Williamson descended from the pulpit, endeavoring to allay the fears of the people. But attention was now called to the fact that several of the congregation were lying in different parts of the church, apparently dead, some of whom had their clothing on fire. Five women were found injured, and having their faces blackened and burned, and a boy had his clothes almost entirely consumed. A respected old parishioner, Mr. Brownlow, aged sixty-eight, was discovered lying at the bottom of his pew, immediately beneath one of the chandeliers, quite dead. There were no marks on the body, but the buttons of his waistcoat were melted, the right

leg of his trousers torn down, and his coat literally burnt off. His wife in the same pew received no injury.”[\[26\]](#)

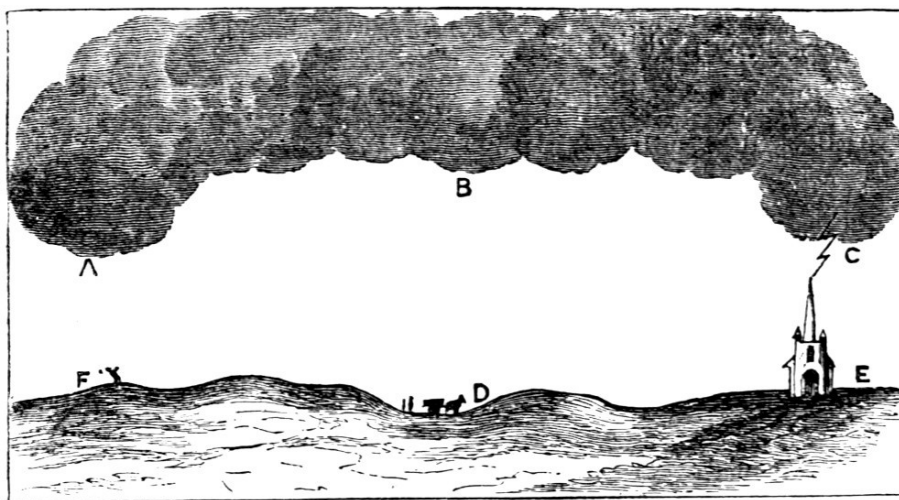


VOLTA'S PISTOL; EXPLOSION CAUSED BY ELECTRIC SPARK.

Not less striking is the story told by Dr. Plummer, surgeon of the Illinois Volunteers, in the *Medical and Surgical Reporter* of June 19, 1865: “Our regiment was yesterday the scene of one of the most terrible calamities which it has been my lot to witness. About two o’clock a violent thunderstorm visited us. While the old guard was being turned out to receive the new, a blinding flash of lightning was seen, accompanied instantly by a terrific peal of thunder. The whole of the old guard, together with part of the new, were thrown violently to the earth. The shock was so severe and sudden that, in most cases, the rear rank men were thrown across the front rank men. One man was instantly killed, and thirty-two men were more or less severely burned by the electric fluid. In some instances the men’s boots and shoes were rent from their feet and torn to pieces, and, strange as it may appear, the men were injured but little in the feet. In all cases the burns appear as if they had been caused by scalding-hot water, in

many instances the skin being shriveled and torn off. The men all seem to be doing well, and a part of them will be able to resume their duties in a few days.”

The Return Shock.—It sometimes happens that people are struck down and even killed at the moment a discharge of lightning takes place between a cloud and the earth, though they are very far from the point where the flash is actually seen to pass; while others, who are situated between them and the lightning, suffer very little, or perhaps not at all. This curious phenomenon was first carefully investigated by Lord Mahon in the year 1779, and was called by him the “return shock.” His theory, which is now commonly accepted, may be easily understood with the aid of the sketch before you.



THE RETURN SHOCK ILLUSTRATED.

Let us suppose ABC to represent the outline of a thundercloud which dips down toward the earth at A and at C. The electricity of the cloud develops by inductive action a charge of the opposite kind in the earth beneath it. But the inductive action is most powerful at E and F, where the cloud comes nearest to the earth. Hence, bodies situated near these points may be very highly electrified as compared with bodies at a point between them, such as D. Now, when a flash of lightning passes at E, the under part of the cloud is at once relieved of its electricity, its inductive action ceases, and, therefore, a person situated at F suddenly ceases to be electrified. This sudden change from a highly electrified to a neutral state involves a shock to his system

which may be severe enough to stun or even to kill him. Meanwhile, people at *D*, having been also electrified to some extent by the influence of the thundercloud, must in like manner undergo a change in their electrical condition when the flash of lightning passes, but this change will be less violent because they were less highly electrified.

Many experiments have been devised to illustrate this theory of Lord Mahon. But the best illustration I know is furnished by this electric machine of Carré's. If you stand near one end of the large conductor when the machine is in action and sparks are taken from the other end, you will feel a distinct electric shock every time a spark passes. The large conductor here takes the place of the cloud, the spark that passes at one end represents the flash of lightning, and the observer at the other end gets the return shock, though he is at a considerable distance from the point where the flash is seen.

An experiment of this kind, of course, cannot be made sensible to a large audience like the present. But I can give you a good idea of the effect by means of this tuft of colored papers. While the machine is in action I hold the tuft of papers near that end of the conductor which is farthest from the point where the discharge takes place. You see the paper ribbons are electrified by induction, and, in virtue of mutual repulsion, stand out from one another "like quills upon the fretful porcupine." But, when a spark passes, the inductive action ceases, the paper ribbons cease to be electrified, and the whole tuft suddenly collapses into its normal state.

While fully accepting Lord Mahon's theory of the return shock as perfectly good so far as it goes, I would venture to point out another influence which must often contribute largely to produce the effect in question, and which is not dependent on the form of the cloud. It may easily happen, from the nature of the surface in the district affected by a thundercloud, that the point of most intense electrification—say *E* in the figure—is in good electrical communication with a distant point, such as *F*, while it is very imperfectly connected with a much nearer point, *D*. In such a case it is evident that bodies at *F* will share largely in the highly-electrified condition of *E*, and also share largely in the sudden change of that condition the moment the flash of lightning passes; whereas bodies at *D* will be less highly electrified before the discharge, and less violently disturbed when the discharge takes place.

This principle may be illustrated by a very simple experiment. Here is a brass chain about twenty feet long. One end of it I hand to any one among the audience who will kindly take hold of it; the other end I hold in my hand. I now stand near the conductor of the machine; and will ask some one to stand about ten feet away from me, near the middle of the chain, but without touching it. Now observe what happens when the machine is worked and I take a spark from the conductor: My friend at the far end of the chain, twenty feet away, gets a shock nearly as severe as the one I get myself, because he is in good electrical communication with the point where the discharge takes place. But my more fortunate friend, who is ten feet nearer to the flash, is hardly sensible of any effect, because he is connected with me only through the floor of the hall, which is, comparatively speaking, a bad conductor of electricity.

Summary.—Let me now briefly sum up the chief destructive effects of lightning. First, with regard to good conductors: though it passes harmlessly through them if they be large enough to afford it an easy passage, it melts and converts them into vapor if they be of such small dimensions as to offer considerable resistance. Secondly, lightning acts with great mechanical force on bad conductors; it is capable of tearing asunder large masses of masonry, and of projecting the fragments to a considerable distance. Thirdly, it sets fire to combustible materials. And lastly, it causes the instantaneous death of men and animals.

Franklin's Lightning Rods.—The object of lightning conductors is to protect life and property from these destructive effects. Their use was first suggested by Franklin, in 1749, even before his famous experiment with the kite; and immediately after that experiment, in 1752, he set up, on his own house, in Philadelphia, the first lightning conductor ever made. He even devised an ingenious contrivance, by means of which he received notice when a thundercloud was approaching. The contrivance consisted of a peal of bells, which he hung on his lightning conductor, and which were set ringing whenever the lightning conductor became charged with electricity.

Franklin's lightning rods were soon adopted in America; and he himself contributed very much to their popularity by the simple and lucid instructions he issued every year, for the benefit of his countrymen, in the annual publication known as "Poor Richard's Almanac." It is very interesting at this distance of time to read the homely practical rules laid

down by this great philosopher and statesman; and, though some modifications have been suggested by the experience of a hundred and thirty years, especially as regards the dimensions of the lightning conductor, it is surprising to find how accurately the general principles of its construction, and of its action, are here set forth.

“It has pleased God,” he says, “in His goodness to mankind, at length to discover to them the means of securing their habitations and other buildings from mischief by thunder and lightning. The method is this: Provide a small iron rod, which may be made of the rod-iron used by nailors, but of such a length that one end being three or four feet in the moist ground, the other may be six or eight feet above the highest part of the building. To the upper end of the rod fasten about a foot of brass wire, the size of a common knitting needle, sharpened to a fine point; the rod may be secured on the house by a few small staples. If the house or barn be long, there may be a rod and point at each end, and a middling wire along the ridge from one to the other. A house thus furnished will not be damaged by lightning, it being attracted by the points and passing through the metal into the ground, without hurting anything. Vessels also having a sharp-pointed rod fixed on the top of their masts, with a wire from the foot of the rod reaching down round one of the shrouds to the water, will not be hurt by lightning.”

Introduction of Lightning Rods into England.—The progress of lightning conductors was more slow in England and on the Continent of Europe, owing to a fear, not unnatural, that they might, in some cases, draw down the lightning where it would not otherwise have fallen. People preferred to take their chance of escaping as they had escaped before, rather than invite, as it were, the lightning to descend on their houses, in the hope that an iron rod would convey it harmless to the earth. But the immense amount of damage done every year by lightning, soon led practical men to entertain a proposal which offered complete immunity from all danger on such easy terms; and when it was found that buildings protected by lightning conductors were, over and over again, struck by lightning without suffering any harm, a general conviction of their utility was gradually established in the public mind.

The first public building protected by a lightning rod in England was St. Paul's Cathedral, in London. On the eighteenth of June, 1764, the beautiful steeple of Saint Bride's Church, in the city, was struck by lightning and

reduced to ruin. This incident awakened the attention of the dean and chapter of St. Paul's to the danger of a similar calamity, which seemed, as it were, impending over their own church. After long deliberation, they referred the matter to the Royal Society, asking for advice and instruction. A committee of scientific men was appointed by the Royal Society to consider the question. Benjamin Franklin himself, who happened to be in London at the time, as the representative of the American States in their dispute with England, was nominated a member of the committee. And the result of its deliberation was that, in the year 1769, a number of lightning conductors were erected on St. Paul's Cathedral.

It was on this occasion that arose the celebrated controversy about the respective merits of points and balls. Franklin had recommended a pointed conductor; but some members of the committee were of opinion that the conductor should end in a ball and not in a point. The decision of the committee was in favor of Franklin's opinion, and pointed conductors were accordingly adopted for St. Paul's Cathedral. But the controversy did not end here. The time was one of great political excitement, and party spirit infused itself even into the peaceful discussions of science. The weight of scientific opinion was on the side of Franklin; but it was hinted, on the other side, that the pointed conductors were tainted with republicanism, and pregnant with danger to the empire. As a rule, the whigs were strongly in favor of points; while the Tories were enthusiastic in their support of balls.

For a time the Tories seemed to prevail. The king was on their side. Experiments on a grand scale were conducted in his presence, at the Pantheon, a large building in Oxford street; he was assured that these experiments proved the great superiority of balls over points; and to give practical effect to his convictions, his majesty directed that a large cannon ball should be fixed on the end of the lightning conductor attached to the royal palace at Kew. But the committee of the Royal Society remained unconvinced. In course of time the heat of party spirit abated; experience as well as reason was found to be in favor of Franklin's views; and the battle of the balls and points has long since passed into the domain of history. [\[27\]](#)

Functions of a Lightning Conductor.—A lightning conductor fulfills two functions. First, it favors a silent and gradual discharge of electricity between the cloud and the earth, and thus tends to prevent that accumulation which must of necessity take place before a flash of lightning

will pass. Secondly, if a flash of lightning come, the lightning conductor offers it a safe channel through which it may pass harmless to the earth.

These two functions of a lightning conductor may be easily illustrated by experiment. When our machine is in action, if I present my closed hand to the large brass conductor, a spark passes between them, and I feel, at the same moment, a slight electric shock. Here the conductor of the machine, as usual, holds the place of the electrified cloud; my closed hand represents, as it were, a lofty building that stands out prominently on the surface of the earth; the spark is the flash of lightning, and the electric shock just suggests the destructive power of the sudden disruptive discharge.

Now let me protect this building by a lightning conductor. For this purpose, I take in my hand a brass rod, which I connect with the earth by a brass chain. In the first instance, I will have a metal ball on the end of my lightning conductor. You see the effect; sparks pass rapidly, but I feel no shock. I can increase the strength of the discharge by hanging this condensing jar on the conductor of the machine. Sparks pass now, much more brilliant and powerful than before, but still I get no shock. It is evident, therefore, that my lightning rod does not prevent the flash from passing, but it conveys it harmless to the ground.

I next take a rod which is sharply pointed, and connecting it as before with the earth by a brass chain, I present the sharp point to the conductor of the machine. Observe how different is the result; there is no disruptive discharge; no spark passes; no shock is felt. Electricity still continues to be generated in the machine, and electricity is generated, by induction, in the brass rod, and in my body. But these two opposite electricities discharge themselves silently, by means of this pointed rod, and no sensible effect of any kind is exhibited.

These experiments are very simple, but they really put before us, in the clearest possible way, the whole theory of lightning conductors. In particular, they give us ocular demonstration that an efficient lightning rod not only makes the lightning harmless when it comes, but tends very much to prevent its coming. A remarkable example, on a large scale, of this important property, is furnished by the town of Pietermaritzburg, the capital of the colony of Natal, in South Africa. This town is subject to the frequent visitation of thunderstorms, at certain seasons of the year, and much damage

was formerly done by lightning, but since the erection of lightning conductors on the principal buildings, the lightning has never fallen within the town. Thunderclouds come as before, but they pass silently over the city, and only begin to emit their lightning flashes when they reach the open country, and have passed beyond the range of the lightning conductors. [\[28\]](#)

But it will often happen, even in the case of a pointed conductor, that the accumulation of electricity goes on so fast that the silent discharge is insufficient to keep it in check. A disruptive discharge will then take place, from time to time, and a flash of lightning will pass. Under these circumstances, the lightning conductor is called upon to fulfill its second function, and to convey the lightning harmless to the earth.

Conditions of a Lightning Conductor.—From the consideration of the functions which it has to fulfill, we may now infer what are the conditions necessary for an efficient lightning conductor. The first condition is that the end of the conductor, projecting into the air, should have, at least, one sharp point. Our experiments have shown us that a pointed conductor tends, in a manner, to suppress the flash of lightning altogether; whereas a blunt conductor, or one ending in a ball, tends only to make it harmless when it comes. It is evident, therefore, that the pointed conductor offers the greater security.

But a fine point is very liable to be melted when the lightning falls upon it, and thus to be rendered less efficient for future service. To meet this danger, it has recently been suggested, by the Lightning Rod Conference, that the extreme end of the conductor should be a blunt point, destined to receive the full force of the lightning flash, when it comes; and that, a little lower down, a number of very fine points should be provided, with a view to favor the silent discharge. This suggestion, which appears admirably fitted to provide for the twofold function of a lightning conductor, deserves to be recorded in the exact terms of the official report.

“It seems best to separate the double functions of the point, prolonging the upper terminal to the very summit, and merely beveling it off, so that, if a disruptive discharge does take place, the full conducting power of the rod may be ready to receive it. At the same time, having regard to the importance of silent discharge from sharp points, we suggest that, at one foot below the extreme top of the upper terminal, there be firmly attached,

by screws and solder, a copper ring bearing three or four copper needles, each six inches long, and tapering from a quarter of an inch diameter to as fine a point as can be made; and with the object of rendering the sharpness as permanent as possible, we advise that they be platinized, gilded, or nickel plated.”^[29]

The second condition of a lightning conductor is, that it should be made of such material, and of such dimensions, as to offer an easy passage to the greatest flash of lightning likely to fall on it; otherwise it might be melted by the discharge, and the lightning, seeking for itself another path, might force its way through bad conductors, which it would partly rend asunder, and partly consume by fire. Copper is now generally regarded as the best material for lightning conductors, and it is almost universally employed in these countries. If it is used in the form of a rope, it should not be less than half an inch in diameter; if a band of copper is preferred—and it is often found more convenient by builders—it should be about an inch and a half broad and an eighth of an inch thick. In France it has been hitherto more usual to employ iron rods for lightning conductors, but since iron is much inferior to copper in its conducting power, the iron rod must be of much larger dimensions; it should be at least one inch in diameter.^[30]

The third condition is that the lightning conductor should be continuous throughout its whole length, and should be placed in good electrical contact with the earth. This is a condition of the first importance, and experience has shown that it is the one most likely of all to be neglected. In a large town the best earth connection is furnished by the system of water-mains and gas-mains, each of which constitutes a great network of conductors everywhere in contact with the earth. Two points, however, must be carefully attended to—first, that the electrical contact between the lightning conductor and the metal pipe should be absolutely perfect; and, secondly, that the pipe selected should be of such large dimensions as to allow the lightning an easy passage through it to the principal main.

If no such system of water-pipes or gas-pipes is at hand, then the lightning rod should be connected with moist earth by means of a bed of charcoal or a metal plate not less than three feet square. This metal plate should be always of the same material as the conductor, otherwise a galvanic action would be set up between the two metals, which in course of time might seriously damage the contact. Dry earth, sand, rock, and shingle

are bad conductors; and, if such materials exist near the surface of the earth, the lightning rod must pass through them and be carried down until it reaches water or permanently damp earth.

Mischief Done by Bad Conductors.—If the earth contact is bad, a lightning conductor does more harm than good. It invites the lightning down upon the building without providing for it, at the same time, a free passage to earth. The consequence is that the lightning forces a way for itself, violently bursting asunder whatever opposes its progress, and setting fire to whatever is combustible.

I will give you some recent and striking examples. In the month of May, 1879, the church of Laughton-en-le-Morthen, in England, though provided with a conductor, was struck by lightning and sustained considerable damage. On examination it was found that the lightning followed the conductor down along the spire as far as the roof; then, changing its course, it forced its way through a buttress of massive masonry, dislodging about two cartloads of stones, and leaped over to the leads of the roof, about six feet distant. It now followed the leads until it came to the cast-iron down-pipes intended to discharge the rain-water, and through these it descended to the earth. When the earth contact of the lightning conductor was examined, it was found exceedingly deficient. The rod was simply bent underground, and buried in dry loose rubbish at a depth not exceeding eighteen inches. This is a very instructive example. The lightning had a choice of two paths—one by the conductor prepared for it, the other by the leads of the roof and the down-pipes—and, by a kind of instinct which, however we may explain, we must always contemplate with wonder, it chose the path of least resistance, though in doing so it had to burst its way at the outset through a massive wall of solid masonry.^[31]

On the 5th of June, in the same year, a flash of lightning struck the house of Mr. Osbaldiston, near Sheffield, and, notwithstanding the supposed protection of a lightning conductor, it did damage to the amount of about five hundred pounds. The lightning here followed the conductor to a point about nine feet from the ground, then passed through a thick wall to a gas-pipe at the back of the drawing-room mirror. It melted the gas-pipe, set fire to the gas, smashed the mirror to atoms, broke the Sevres vases on the chimney-piece, and dashed the furniture about. In this case, as in the former, it was found that the earth contact was bad; and, in addition, the

conductor itself was of too small dimensions. Hence, the electric discharge found an easier path to earth through the gas-pipes, though to reach them it had to force for itself a passage through a resisting mass of non-conductors. [32]

Again in the same year, on the 28th of May, the house of Mr. Tomes, of Caterham, was struck by lightning, and some slight damage was done. After a careful examination it was found that the greater part of the discharge left the lightning conductor with which the house was provided, and passed over the slope of the roof to an attic room, into which it forced its way through a brick wall, and reached a small iron cistern. This cistern was connected by an iron pipe of considerable dimensions with two pumps in the basement story; and through them the lightning found an easy passage to the earth, and did but little harm on its way. When the earth contact of the lightning conductor was examined, it was discovered that the end of the rod was simply stuck into a dry chalky soil to a depth of about twelve inches. Thus in this case, as in the two former, it was made quite clear that the lightning conductor failed to fulfill its functions because the earth contact was bad. [33]

Cases are not uncommon in which builders provide underground a carefully constructed reservoir of water, into which the lower end of the lightning rod is introduced. The idea seems to prevail that a reservoir of water constitutes a good earth contact; and this is quite true of a natural reservoir, such as a lake, where the water is in contact with moist earth over a considerable area. But an artificial reservoir may have quite an opposite character, and practically insulate the lightning conductor from the earth. One which came under my notice lately, in the neighborhood of this city, consists of a large earthenware pipe set on end in a bed of cement, and kept half full of water. Now, the earthenware pipe is a good insulator, and so is the bed of cement in which it rests; and the whole arrangement is identical, in all essential features, with the apparatus of Professor Richman, in which he introduced his lightning rod into a glass bottle, and by which he lost his life a hundred and thirty years ago.

A conductor mounted in this manner will, probably enough, draw down lightning from the clouds; but it is more likely to discharge it, with destructive effect, into the building it is intended to guard, than to transmit it harmlessly to the earth. An example is at hand in the case of Christ

Church, in the town of Clevedon, in Somersetshire. This church was provided with a very efficient system of lightning conductors, five in number, corresponding to the four pinnacles and the flagstaff, on the summit of the principal tower. The five conductors consisted of good copper-wire rope; all were united together inside the tower, through which they were carried down to earth, and there ended in an earthenware drain. This kind of earth contact might be pretty good as long as water was flowing in the drain; but whenever the drain was dry the conductor was practically insulated from the earth. On the fifteenth of March, 1876, the church was struck by lightning, which for some distance followed the line of the conductor; then finding its passage barred by the earthenware drain, which was dry at the time, it burst through the walls of the church, displacing several hundredweight of stone, and making its way to earth through the gas-pipe.^[34]

Another very instructive example is furnished by the lightning conductor attached to the lighthouse of Berehaven, on the south-west coast of Ireland. It consists of a half-inch copper-wire rope, which is carried down the face of the tower “until it reaches the rock at its base, where it terminates in a *small hole, three inches by three inches, jumped out of the rock, about six inches under the surface.*” Here, again, we have a good imitation of Professor Richman’s experiment, with only this difference, that a small hole in the rock is substituted for a glass bottle. A lightning conductor of this kind fulfills two functions: it increases the chance of the lightning coming down on the building, and it makes it positively certain that, having come, it cannot get to earth without doing mischief.

The lightning did come down on the Berehaven Lighthouse, about five years ago. As might have been expected, it made no use of the lightning conductor in finding a path to earth, but forced its way through the building, dealing destruction around as it descended from stage to stage. The Board of Irish Lights furnished a detailed report of this accident to the Lightning Rod Conference, in March, 1880, from which the above particulars have been derived.^[35]

Precaution Against Rival Conductors.—But it is not enough to provide a good lightning conductor, which is itself able to convey the electric discharge harmless to the earth; we must take care that there are no rival conductors near at hand in the building, to draw off the lightning from the

path prepared for it, and conduct it by another route in which its course might be marked with destruction. This precaution is of especial importance at the present day, owing to the great extent to which metal, of various kinds, is employed in the construction and fittings of modern buildings. I will take a typical case which will bring home this point clearly to your minds.

A great part of the roof of many large buildings is covered with lead. The lead, at one or more points may come near the gutters intended to collect the rain water; the gutters are in connection with the cast-iron down-pipes into which the water flows, and these down-pipes often pass into the earth, which, under the circumstances, is generally moist, and, therefore, in good electrical contact with the metal pipes. Here, then, is an irregular line of conductors, which, though it has gaps here and there, may, under certain conditions, offer to the lightning discharge a path not less free than the lightning conductor itself. What is the consequence? The flash of lightning, or a part of it, will quit the lightning rod, and make its way to earth through the broken series of conductors, doing, perhaps, serious mischief, as it leaps across, or bursts asunder, the non-conducting links in the chain.

Another illustration may be taken from the gas and water-pipes, with which almost all buildings in great cities are now provided, and which constitute a network of conductors, spreading out over the walls and ceilings, and stretching down into the earth, with which they have the best possible electrical contact. Now, it often happens that a lightning conductor, at some point in its course, comes within a short distance of this network of pipes. In such a case, a portion of the electrical discharge is apt to leave the lightning conductor, force its way destructively through masses of masonry, enter the network of pipes, melt the leaden gas-pipe, ignite the gas, and set the building on fire.

These are not merely the speculations of philosophers. All the various incidents I have just described have occurred, over and over again, during the last few years. You will remember, in some of the examples I have already set before you, when the electric discharge failed to find a sufficient path to earth through the lightning rod, it followed some such broken series of chance conductors as we are now considering. But this broken series of conductors seems to bring with it a special danger of its own, even when the

lightning conductor is otherwise in efficient working order. I will give you just one case in point.

On the fifth of June, 1879, the Church of Saint Marie, Rugby, was struck by lightning and set on fire, and narrowly escaped being burned to the ground. A number of workmen were engaged on that day in repairing the spire of the church. About three o'clock they saw a dense black cloud approaching, and they came down to take shelter within the building. In a few minutes they heard a terrific crash just overhead; at the same moment the gas was lighted under the organ loft and the woodwork was set in a blaze. The men soon succeeded in putting out the fire, and the church escaped with very little damage.

Now, in this case there was no reason to suppose that the lightning conductor was in any way defective. But about half-way up the spire there was a peal of eight bells. Attached to these bells were iron wires, about the eighth of an inch in diameter, leading from the clappers down to the organ-loft, where they came within a short distance of a gas-pipe fixed in the wall. It would seem that a great part of the discharge was carried safely to earth by the lightning conductor. But a part branched off at the bells in the spire, descended by the iron wires, and forced its way into the organ loft, to reach the network of gas-pipes, through which it passed down to the earth, melting the soft leaden gas-pipe in its course and lighting the gas.

The remedy for this danger is obvious. All large masses of metal used in the structure of a building—the leads and gutters of the roof, the cast-iron down-pipes, the iron gas and water mains—should be put in good metallic connection with the lightning conductor, and, as far as may be, with one another. Connected in this way they furnish a continuous and effective line of conductors leading safely down to earth; and, instead of being a dangerous rival, they become a useful auxiliary to the lightning rod.

I would observe, however, that the lightning conductor ought not to be connected directly with the soft leaden pipes which are commonly employed to convey gas and water to the several parts of a building. Such pipes, as we have seen, are liable to be melted when any considerable part of the lightning discharge passes through them; and thus much harm might be done, and the building might even be set on fire by the lighting of the gas. Every good end will be attained if the conductor is put in metallic

connection with the iron gas and water *mains* either inside or outside the building.

Insulation of Lightning Conductors.—It is a question often asked whether a lightning rod should be insulated from the building it is intended to protect. I believe that this practice was formerly recommended by some writers, and I have observed that glass insulators are still employed not infrequently by builders in the erection of lightning conductors; but, from the principles I have set before you to-day, it seems clear that any insulation of this kind is, to say the least, altogether useless. The building to be protected is itself in electrical communication with the earth, and the lightning conductor, if efficient, is also in electrical communication with the earth—therefore, the lightning conductor and the building are in electrical communication with each other through the earth, and any attempt at insulating them from one another above the earth is only labor thrown away.

Further, I have just shown you that the masses of metal employed in the structure or decoration of a building ought to be electrically connected with each other and with the lightning conductor. Now, if this be done, the lightning conductor is, by the fact, in direct communication with the building, and the glass insulators are utterly futile. Again, the building itself, during a thunderstorm, becomes highly electrified by the inductive action of the cloud, and needs to be discharged through the conductor just as the surrounding earth needs to be discharged; therefore, the more thoroughly it is connected with the conductor, the more effectively will the conductor fulfill its functions.

Personal Safety in a Thunderstorm.—I suppose there is hardly any one to whom the question has not occurred, at some time or another, what he had best do to secure his personal safety during a thunderstorm. This question is of so much practical interest that I think I shall be excused if I say a few words about it, though perhaps, strictly speaking, it is somewhat beside the subject of lightning conductors.

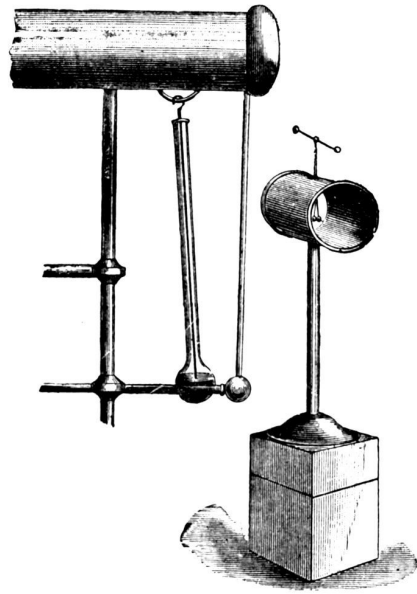
At the outset, perhaps, I shall surprise you when I say that you would enjoy the most perfect security if you were in a chamber entirely composed of metal plates, or in a cage constructed of metal bars, or if you were incased, like the knights of old, in a complete suit of metal armor. This kind

of defense is looked upon as so perfect, among scientific men, that Professor Tait does not hesitate to recommend his adventurous young friends devoted to the cause of science to provide themselves with a light suit of copper, and, thus protected, take the first opportunity of plunging into a thundercloud, there to investigate, at its source, the process by which lightning is manufactured.^[36]

The reason why a metal covering affords complete protection is that, when a conductor is electrified, the whole charge of electricity exists on the outside surface of the conductor; and therefore, when a discharge takes place, it is only the outside surface that is affected. Thus, if you were completely incased in a metal covering, and then charged with electricity by the inductive action of a thundercloud, it is only the metal covering that would undergo any change of electrical condition; and when the lightning flash would pass, it is only the metal covering that would be discharged.

Let me show you a very pretty and interesting experiment to illustrate this principle: Here is a hollow brass cylinder, open at the ends, mounted on an insulating stand. On the outside is erected a light brass rod with two pith balls suspended from it by linen threads. Two pith balls are also suspended by linen threads from the inner surface of the cylinder. You know that these pith balls will indicate to us the electrical condition of the surfaces to which they are attached. If the surface be electrified, the pith balls attached to it will share in its electrical condition, and will repel each other; if the surface be neutral, the pith balls attached to it will be neutral, and will remain at rest.

I now put this apparatus under the influence of our thundercloud, that is, the large brass conductor of our machine. The moment my assistant turns the handle, the electricity begins to be developed on the conductor, and you see, at once, the effect on the brass cylinder. The pith balls attached to the outer surface fly asunder; those attached to the inner surface remain at rest. And now a spark passes; our thundercloud is discharged; the inductive action ceases; the pith balls on the outside suddenly collapse, while those on the inside are in no way affected.



PROTECTION FROM LIGHTNING FURNISHED BY A CLOSED CONDUCTOR.

It is not necessary that the brass cylinder should be insulated. To vary the experiment, I will now connect it with the earth by a chain; you will observe that the effect is precisely the same as before. Flash after flash passes while the machine continues in action; the outside pith balls fly about violently, being charged and discharged alternately; the inside pith balls remain all the time at rest. Thus you see clearly that, if you were sitting inside such a metal chamber as this, or covered with a complete suit of metal armor, you would be perfectly secure during a thunderstorm, whether the chamber were electrically connected with the earth or insulated from it.

Practical Rules.—But it rarely happens, when a thunderstorm comes, that an iron hut or a complete suit of armor is at hand, and you will naturally ask me what you ought to do under ordinary circumstances. First, let me tell you what you ought not to do. You ought not to take shelter under a tree, or under a haystack, or under the lee of a house; you ought not to stand on the bank of a river, or close to a large sheet of water. If indoors, you ought not to stay near the fireplace, or near any of the flues or chimneys; you ought not to stand under a gasolier hanging from the ceiling; you ought not to remain close to the gas pipes or water-pipes, or any large

masses of metal, whether used in the construction of the building, or lying loosely about.

The necessity for these precautions is sufficiently evident from the principles I have already put before you. You want to prevent your body from becoming a link in that broken chain of conductors which, as we have seen, the electric discharge between earth and cloud is likely to follow. Now a tree is a better conductor than the air; and your body is a better conductor than a tree. Hence, the lightning, in choosing the path of least resistance, would leave the air to pass through the tree, and would leave the tree to pass through you. A like danger would await you if you stood under the lee of a haystack or of a house.

The number of people who lose their lives by taking refuge under trees in thunderstorms is very remarkable. As one instance out of many, I may cite the following case which was reported in the *Times*, July 14, 1887: "Yesterday the funeral of a negress was being conducted in a graveyard at Mount Pleasant, sixty miles north of Nashville, Tennessee, when a storm came on, and the crowd ran for shelter under the trees. Nine persons stood under a large oak, which the lightning struck, killing everyone, including three clergymen, and the mother and two sisters of the girl who had been buried."

Again, every large sheet of water constitutes practically a great conductor, which offers a very perfect medium of discharge between the earth round about and the cloud. Therefore, when a thundercloud is overhead, the sheet of water is likely to become one end of the line of the lightning discharge; and if you be standing near it, the line of discharge may pass through your body.

When lightning strikes a building, it is very apt to use the stack of chimneys in making its way to earth, partly because the stack of chimneys is generally the most prominent part of the building, and partly because, on account of the heated air and the soot within the chimney, it is usually a moderately good conductor. Therefore, if you be indoors, you must keep well away from the chimneys; and for a similar reason, you must keep as far as you can from large masses of metal of every kind.

Having pointed out the sources of danger which you must try to avoid in a thunderstorm, I have nearly exhausted all the practical advice that I have

at my command. But there are some occasions on which it may be possible, not only to avoid evident sources of danger, but to make special provision for your own security. Thus, for example, in the open country, if you stand a short distance from a wood, you may consider yourself as practically protected by a lightning conductor. For a wood, by its numerous branches and leaves, favors very much a quiet discharge of electricity, thus tending to suppress altogether the flash of lightning; and if the flash of lightning does come, it is much more likely to strike the wood than to strike you, because the wood is a far more prominent body, and offers, on the whole, an easier path to earth. In like manner, if you place yourself near a tall solitary tree, some twenty or thirty yards outside its longest branches, you will be in a position of comparative safety. If the storm overtake you in the open plain, far away from trees and buildings, you will be safer lying flat on the ground than standing erect.

In an ordinary dwelling house, the best situation is probably the middle story, and the best position in the room is in the middle of the floor; provided, of course, that there is no gasolier hanging from the ceiling above or below you. Strictly speaking, the *middle of the room* would be a still safer position than the middle of the floor; and nothing could be more perfect than the plan suggested by Franklin, to get into “a hammock, or swinging bed, suspended by silk cords, and equally distant from the walls on every side, as well as from the ceiling and floor, above and below.” An interesting case has been recently recorded, by a resident of Venezuela, which illustrates in a remarkable way the excellence of this advice. “The lightning,” he says, “struck a *rancho*—a small country house, built of wood and mud, and thatched with straw or large leaves—where one man slept in a hammock, another lay under the hammock on the ground, and three women were busy about the floor; there were also several hens and a pig. The man in the hammock did not receive any injury whatever, while the other four persons and the animals were killed.”^[37]

But, as I can hardly hope that many of you when the thunderstorm actually comes will find yourselves provided with a hammock, I would recommend, as more generally useful, another plan of Franklin’s, which is simply to sit on one chair in the middle of the floor and put your feet up on another. This arrangement will approach very nearly to absolute security if

you take the further precaution, also mentioned by Franklin, of putting a feather bed or a couple of hair mattresses under the chairs. [\[38\]](#)

Security Afforded by Lightning Rods.—You might, perhaps, be inclined to infer hastily, from the examples I have set before you, in the course of this lecture, of buildings which were struck and severely injured by lightning though provided with lightning conductors, that a lightning rod affords a very imperfect protection to life and property. But such an idea would be entirely at variance with the evidence at hand on the subject. In all the cases to which I have referred, and in many others which might easily have been cited, the damage was done simply because the lightning rods were deficient in one or more of the conditions on which I have so much insisted. Where these conditions are fulfilled, the lightning flash will either not come down at all upon the building, or, if it do come, it will be carried harmless to the earth.

Perhaps there is no one fact that so forcibly brings home to the mind the complete protection afforded by lightning conductors as the change which followed their introduction into the Royal Navy. I have already told you that in former times the damage done by lightning to ships of the Royal Navy was a regular source of expenditure, amounting every year to several thousand pounds sterling. But, after the general adoption of lightning conductors about forty years ago, through the indefatigable exertions of Sir William Snow Harris, this source of expenditure absolutely disappeared, and injury to life and property has long been practically unknown in Her Majesty's Fleet.

I should say, however, that the trial of lightning conductors in the Navy, though it lasted long enough to prove their perfect efficiency, has almost come to an end in our own days. The great iron monsters which in recent times have taken the place of the wooden ships of Old England are quite independent of lightning rods in the common sense of the word. Their ponderous masts are virtually lightning rods of colossal dimensions, and their unsightly hulls are, so to speak, earth-plates of enormous size in perfect electrical contact with the ocean. To add to such structures lightning conductors of the common kind would be nothing better than “wasteful and ridiculous excess.”

As regards buildings on land, I may refer to the little province of Schleswig-Holstein, of which I have already spoken to you. From some cause or other this small peninsula is singularly exposed to thunderstorms, and of late years it has been more abundantly provided with lightning conductors than, perhaps, any other district of equal extent in Europe. Now, as a simple illustration of the protection afforded by these lightning conductors, I may mention that, on the 26th of May, 1878, a violent thunderstorm burst over the little town of Utersen. Five several flashes of lightning fell in different parts of the town, but not the slightest harm was done, each flash being safely carried to earth by a lightning conductor. Further, it appears from the records of the fire insurance company that, out of 552 buildings injured by lightning during a period of eight years—from 1870 to 1878—only four had lightning conductors; and in these four cases it was found, on examination, that the lightning conductors were defective. [\[39\]](#)

It would be easy to multiply evidence on this subject. But as I have already trespassed, I fear, too far on your patience, I will content myself with saying, in conclusion, that according to all the highest authorities, both practical and theoretical, any structure provided with a lightning conductor properly fitted up in conformity with the principles I have set before you to-day is perfectly secure against lightning. The lightning, indeed, may fall upon it, but it will pass harmless to the earth; and the experience of more than a hundred years has fully justified the simple and modest words of the great inventor of lightning conductors: “It has pleased God, in His goodness to mankind, at length to discover to them the means of securing their habitations and other buildings from mischief by thunder and lightning.”

NOTE I.

ON THE LIGHTNING CONDUCTOR AT BEREHAVEN. [\[40\]](#)

It is satisfactory to know that the lightning conductor referred to in my lecture as attached to the lighthouse at Berehaven has been put in good order under the best scientific guidance. The following interesting letter from Professor Tyndall, which appeared in the

Times, August 31, 1887, gives the history of the matter very clearly, and fully bears out the views put forward in my lecture:

“Your recent remarks on thunderstorms and their effects induce me to submit to you the following facts and considerations. Some years ago a rock lighthouse on the coast of Ireland was struck and damaged by lightning. An engineer was sent down to report on the occurrence; and, as I then held the honorable and responsible post of scientific adviser to the Trinity House and Board of Trade, the report was submitted to me. The lightning conductor had been carried down the lighthouse tower, its lower extremity being carefully embedded in a stone perforated to receive it. If the object had been to invite the lightning to strike the tower, a better arrangement could hardly have been adopted.

“I gave directions to have the conductor immediately prolonged, and to have added to it a large terminal plate of copper, which was to be completely submerged in the sea. The obvious convenience of a chain as a prolongation of the conductor caused the authorities in Ireland to propose it; but I was obliged to veto the adoption of the chain. The contact of link with link is never perfect. I had, moreover, beside me a portion of a chain cable through which a lightning discharge had passed, the electricity in passing from link to link encountering a resistance sufficient to enable it to partially fuse the chain. The abolition of resistance is absolutely necessary in connecting a lightning conductor with the earth, and this is done by closely embedding in the earth a plate of good conducting material and of large area. The largeness of area makes atonement for the imperfect conductivity of earth. The plate, in fact, constitutes a wide door through which the electricity passes freely into the earth, its disruptive and damaging effects being thereby avoided.

“These truths are elementary, but they are often neglected. I watched with interest some time ago the operation of setting up a lightning conductor on the house of a neighbor of mine in the country. The wire rope which formed part of the conductor was carried down the wall and comfortably laid in the earth below without any terminal plate whatever. I expostulated with the man who did the work, but he obviously thought he knew more about the

matter than I did. I am credibly informed that this is a common way of dealing with lightning conductors by ignorant practitioners, and the Bishop of Winchester's palace at Farnham has been mentioned to me as an edifice 'protected' in this fashion. If my informant be correct, the 'protection' is a mockery, a delusion, and a snare."

NOTE II.

BOOKS OF REFERENCE.

As some of my readers may wish to pursue the study of lightning and lightning conductors beyond the limits to which a popular lecture must, of necessity, be confined, I subjoin a list of the books which I think they would be likely to find most useful for the purpose. Among ordinary text-books on physics—Jamin, *Cours de Physique*, vol. i., pp. 470-494; Mascart, *Traité d'Electricité Statique*, vol. ii., pp. 555-579; De Larive, *A Treatise on Electricity*, in three volumes, London, 1853-8, vol. iii., pp. 90-201; Daguin, *Traité de Physique*, vol. iii., pp. 209-280; Riess, *Die Lehre von der Reibungs-Elektricität*, vol. ii., pp. 494-564; Müller-Pouillet, *Lehrbuch der Physik*, Braunschweig, 1881, vol. iii., pp. 210-225; Scott, *Elementary Meteorology*, chap. x. Of the numerous special treatises and detached papers on the subject, I would recommend *Instruction sur les Paratonnerres adopté par l'Académie des Sciences*, Part i., 1823, Part ii., 1854, Part iii., 1867, Paris, 1874; Arago, *Sur le Tonnerre*, Paris, 1837; also his *Meteorological Essays*, translated by Sabine, London, 1855; Sir William Snow Harris, *On the Nature of Thunderstorms*, London, 1843; also by the same writer, *A Treatise on Frictional Electricity*, London, 1867; and various papers on lightning conductors, from 1822 to 1859; Tomlinson, *The Thunderstorm*, London, 1877; Anderson, *Lightning Conductors*, London, 1880; Holtz, *Ueber die Theorie, die Anlage, und die Prüfung der Blitzableiter*, Greifswald, 1878; Weber, *Berichte über Blitzschläge in der Provinz Schleswig-Holstein*, Kiel, 1880-1; Tait, *A Lecture on Thunderstorms*, delivered in the City Hall, Glasgow, in 1880, *Nature*, vol. xxii.; Report of the Lightning Rod Conference,

London, 1882. This last-mentioned volume comes to us with very high authority, representing, as it does, the joint labors of several eminent scientific men selected from the following societies: The Meteorological Society, the Royal Institute of British Architects, the Society of Telegraph Engineers and Electricians, the Physical Society.

Since the above was in print, two lectures given before the Society of Arts by Professor Oliver Lodge, F. R. S., have appeared in the *Electrician*, June and July, 1888, in which some new views are put forward respecting lightning conductors, that seem deserving of careful consideration.

FOOTNOTES:

[17] The Thunderstorm, by Charles Tomlinson, F. R. S., Third Edition, pp. 153-4.

[18] Two Lectures on Atmospheric Electricity and Protection from Lightning, published at the end of his Treatise on Frictional Electricity, p. 273.

[19] See Report of Lightning Rod Conference, p. 119.

[20] *Loco citato*.

[21] Sir William Snow Harris, *loco citato*, p. 274.

[22] *Id.*, p. 275.

[23] The Thunderstorm, by Charles Tomlinson, F.R.S., Third Edition, p. 172.

[24] See for these facts, Anderson, Lightning Conductors, p. 197; Tomlinson, The Thunderstorm, pp. 167-9; Harris, *loco citato*, pp. 273-4.

[25] See Anderson, Lightning Conductors, pp. 170-5.

[26] The Thunderstorm, pp. 158-9. See also an account of four persons who were struck on the Matterhorn, in July, 1869, all of whom were hurt, and none killed: Whymper's Scrambles Among the Alps, pp. 414, 415.

[27] See Philosophical Transactions of the Royal Society, 1773, p. 42, and 1778, part i., p. 232; Anderson's Lightning Conductors, pp. 40-2; Lightning Rod Conference, pp. 76-9.

[28] See A Lecture on Thunderstorms, by Professor Tait of Edinburgh, published in Nature, vol. xxii., p. 365.

[29] Report of the Lightning Rod Conference, p. 4.

[30] The dimensions here set forth are greater in some respects than those "recommended as a minimum" in the report of the Lightning Rod Conference, page 6. But it will be observed by those who consult the report that the minimum

recommended is just the size which, in the preceding paragraph of the report, is said to have been actually melted by a flash of lightning; and, therefore, it seems not to be a very safe minimum. It will be also seen that there is some confusion in the figures given, and that they contradict one another. For the dimensions of iron rods, see the instructions adopted by the Academy of Science, Paris, May 20, 1875; Lightning Rod Conference, pp. 67-8.

[31] See letter of Mr. R. S. Newall, F. R. S., in the *Times*, May 30, 1879.

[32] See *Nature*, June 12, 1879, vol. xx., p. 146.

[33] See letter of Mr. Tomes in *Nature*, vol. xx., p. 145; also Lightning Rod Conference, pp. 210-15.

[34] See Anderson, *Lightning Conductors*, pp. 208-10.

[35] See Lightning Rod Conference, pp. 208-10; see also the note at the end of this Lecture, p. 52.

[36] Lecture on Thunderstorms, *Nature*, vol. xxii., pp. 365, 437. See, also, a very interesting paper by the late Professor J. Clerk Maxwell, read before the British Association at Glasgow in 1876, and reprinted in the report of the Lightning Rod Conference, pp. 109, 110.

[37] *Nature*, vol. xxxi., p. 459.

[38] See further information on this interesting subject in the Report of the Lightning Rod Conference, pp. 233-5.

[39] See "Die Theorie, die Anlage, und die Prüfung der Blitzableiter," von Doctor W. Holtz, Griefswald, 1878.

[40] See [page 44](#).

APPENDIX.

RECENT CONTROVERSY ON LIGHTNING CONDUCTORS.

The lecture on lightning conductors contained in this volume fairly represents, I think, the theory hitherto received on the subject. It is, moreover, entirely in accord with the report of the Lightning Rod Conference, brought out in 1883, by a committee of most eminent men, representing several branches of science, who were specially chosen to consider this question some ten years ago.

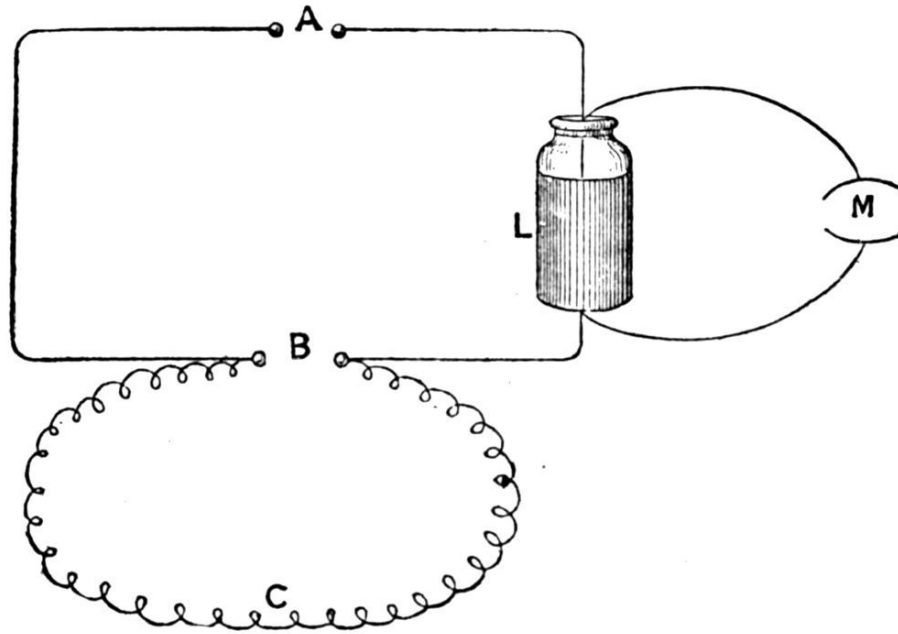
Lectures of Professor Lodge.—But, in the month of March, 1888, two lectures were given before the Society of Arts, in London, by Professor Oliver Lodge, in which this theory was directly challenged, and attacked with cogent arguments, supported by striking and original experiments. These lectures gave rise to an animated controversy, which culminated in a formal discussion at the recent meeting of the British Association in Bath. The discussion was carried on with great spirit, and most of the leading representatives of physical and mechanical science took an active part in it. The greater portion of this volume was printed off before the meeting of the British Association took place. But the discussion on the theory of lightning conductors seemed to me so interesting and important that I thought it right, in the form of an Appendix, to give some account of the questions at issue, and of the opinions expressed upon them.

Professor Lodge maintains^[41] that the received theory of lightning rods is open to two objections. First, it takes account only of the conducting power of the lightning rod, and takes no account of the phenomenon known as self-induction, or electrical inertia. Secondly, it assumes that the whole substance of a lightning rod acts as a conductor, in all cases of lightning discharge; whereas there is reason to believe that, in many cases, it is only a thin outer shell that really comes into action. I will deal with these two points separately.

The Effect of Self-Induction.—When an electric discharge begins to pass through a conductor, a momentary back electro-motive force is developed in the conductor, which obstructs its passage. This phenomenon is called by some self-induction, by others electrical inertia; but its existence is admitted by all. Now, when a flash of lightning, so to say, falls on a lightning rod, the back electro-motive force developed is very considerable; and it may offer so great an obstruction that the discharge will find an easier passage by some other route, such as the stone walls and woodwork, and furniture of the building.

According to this view, the obstruction which a flash of lightning encounters in a conductor consists partly of the resistance of the conductor, in the ordinary sense of the word resistance, and partly of the back electro-motive force due to self-induction. The sum of these two Professor Lodge calls the *impedance* of the lightning rod; and he considers that the impedance may be enormously great, even when the resistance, in the ordinary sense, is comparatively small.

In support of this view he has devised the following extremely ingenious and remarkable experiment. A large Leyden jar, L, was arranged in such a manner that, while it received a steady charge from an electrical machine, it discharged itself, at intervals, across the air space at A, between two brass balls. The discharge had then two alternative paths before it; one through a conducting wire, C, the other across a second air space, between two brass balls at B. During the experiment, the two balls at A were kept at a fixed distance of one inch apart; but the distance between the two balls at B was varied. The conductor, C, used in the first instance, was a stout copper wire, about forty feet long, and having a resistance of only one-fortieth of an ohm.



INDUCTION EFFECT OF LEYDEN JAR DISCHARGE.

M Electrical Machine.

L Leyden Jar.

A B Air Spaces between Brass Knobs.

C Conducting Wire.

It was found that, so long as the distance between the B knobs was less than 1.43 inches, all the discharges passed across between the knobs, in the form of a spark. When the distance exceeded 1.43 inches, all the discharges passed through the conductor, C, and no spark appeared between the balls at B. And when the distance was exactly 1.43 inches, the discharge sometimes took place between the knobs, and sometimes followed the conductor, C. The interpretation given to these facts is that the obstruction offered by the conductor C was about equal to the resistance of 1.43 inches of air; and it is proposed to call this distance, under the conditions of the experiment, the *critical distance*.

Coming now to the application of these results, Professor Lodge argues that the conductor C, in his experiment, represents a lightning rod of unimpeachable excellence; and yet, in certain cases, the discharge refuses to follow the conductor, and prefers to leap across a considerable space of air, notwithstanding the enormous resistance it there encounters. In like manner, he says, a flash of lightning may, in certain cases, leave a lightning rod fitted up in the most orthodox manner, and force its way to earth through

resisting masses of mason work and such chance conductors as may come across its path.

This conclusion, he admits, is altogether at variance with the received views on the subject; but he contends that it is perfectly in accord with the scientific theory of an electrical discharge. The moment the discharge begins to pass in the conductor, it encounters the obstruction due to self-induction; and this obstruction is so great that the bad conductors offer, on the whole, an easier path to earth.

Variation of the Experiment.—When the experiment was varied by substituting a thin iron wire for the stout copper wire at first employed, a very curious result was obtained. The wire chosen was of the same length as the copper, but had a resistance about 1,300 times as great; its resistance being, in fact, 33.3 ohms. Nevertheless, in this experiment, when the B knobs were at a distance of 1.43 inches, no spark passed, which showed that the discharge always followed the line of the conductor, and therefore that the conductor offered less obstruction than 1.43 inches of air. The knobs were then brought gradually nearer and nearer; and it was not until the distance was considerably reduced that the sparks began to pass between them. When the distance was exactly 1.03 inches, the discharge sometimes passed between the knobs, and sometimes through the conductor; this was, therefore, the *critical distance*, in the case of the iron wire. Thus it appeared that the obstruction offered to the discharge by the iron wire was much less than that offered by the copper, the one being equal to a resistance of only 1.03 inches of air, the other to a resistance of 1.43 inches.

It does not appear that Professor Lodge undertakes to offer any satisfactory explanation of this result. He has come to the conclusion, from his various experiments, that, in the case of a sudden discharge, difference of conducting power between fairly good conductors is a matter of practically no account; and that difference of sectional area is a matter of only trifling account. But he does not see why a thin iron wire should have a *smaller* impedance than a much thicker wire of copper. He proposes to repeat the experiments so as to confirm or to modify the result, which for the present seems to him anomalous. ^[42]

The Outer Shell only of a Lightning Rod Acts as a Conductor.—As a consequence of self-induction or electrical inertia, Professor Lodge

contends that a lightning discharge in a conductor consists of a series of oscillations. These oscillations follow one another with extraordinary rapidity—there may be a hundred thousand in a second, there may be a million. Now it has been shown that, when a current starts in a conductor, it does not start at once all through its section; it begins on the outside, and then gradually, but rapidly, penetrates to the interior. From this he infers that the extremely rapid oscillations of a lightning discharge have not time to penetrate to the interior of a conductor. The electricity keeps surging to and fro in the superficial layer or outer shell, while the interior substance of the rod remains inert and takes no part in the action. A conductor, therefore, will be most efficient for carrying off a flash of lightning if it present the greatest possible amount of surface; a thin, flat tape will be more efficient than a rod of the same mass; and a number of detached wires more efficient than a solid cylinder. As for existing lightning conductors, the greater part of their mass would, in many cases, have no efficacy whatever in carrying off a flash of lightning.

The Discussion.—The discussion at the meeting of the British Association was opened by Mr. William H. Preece, F.R.S., Electrician to the Post Office, who claimed to have 500,000 lightning conductors under his control. He expressed his conviction that a lightning rod, properly erected and duly maintained, was a perfect protection against injury from lightning; and in support of this conviction he urged very strongly the report of the Lightning Rod Conference. This report represented the mature judgment of the most eminent scientific men, who had devoted years to the study of the question; and he wished particularly to bring before the meeting their clear and decisive assertion—an assertion he was there to defend—that “there is no authentic case on record where a properly constructed conductor failed to do its duty.”

The new views put forward by Professor Lodge were based, in great measure, on his theory that a lightning discharge consisted of a series of rapid oscillations. But this theory should be received with great caution. It seemed to be nothing more than a deduction from certain mathematical formulas, and was not supported by any solid basis of observation or experiment. Besides, there were many facts against it. They all knew that a flash of lightning magnetized steel bars, deranged the compasses of ships at sea, and transmitted signals on telegraph wires. But such effects could not

be produced by a series of oscillations, which, being equal and opposite, would neutralize each other. It was alleged that these rapid oscillations occurred in the discharge of a Leyden jar. That might be true, and probably was true; but they were not dealing with Leyden jars, they were dealing with flashes of lightning. If there was any analogy between the discharge of a Leyden jar and a flash of lightning, it was to be found, not in the external discharge employed by Professor Lodge in his experiments, but in the bursting of the glass cylinder between the two coatings of the jar.

Lord Rayleigh thought the experiments of Professor Lodge were likely to have important practical applications to lightning conductors. But though these experiments were valuable as suggestions, they did not furnish a sufficient ground for adopting any new system of protection. It was only by experience with lightning conductors themselves that the question could be finally settled.

Sir William Thomson hoped for great fruit from the further investigation of self-induction in the case of sudden electrical discharges. He warmly encouraged Professor Lodge to continue his researches; but he expressed no decided opinion on the question at issue. Incidentally he observed that the best security for a gun-powder magazine was an iron house; no lightning conductor at all, but an iron roof, iron walls, and an iron floor. Wooden boards should, of course, be placed over the floor to prevent the danger of sparks from people walking on sheet-iron. This iron magazine might be placed on a dry granite rock, or on wet ground; it might even be placed on a foundation under water; it might be placed anywhere they pleased; no matter what the surroundings were, the interior would be safe. He thought that was an important practical conclusion which might safely be drawn from the consideration of these electrical oscillations and the experiments regarding them.

Professor Rowland, of the Johns Hopkins University, America, said that the question seemed to be whether the experiment of Professor Lodge actually represented the case of lightning. He was very much disposed to think it did not. In the experiment almost the whole circuit consisted of good conductors; whereas, in the case of lightning, the path of the discharge was, for the most part, through the air, and therefore it might be an entirely different phenomenon. The air being a very bad conductor, a flash of lightning might, perhaps, not consist of oscillations, but rather of a single

swing. Moreover, it was not at all clear that the length of the spark, in the experiment, could be taken as a measure of the obstruction offered by the conductor. Professor George Forbes was greatly impressed with the beauty and significance of Professor Lodge's experiments, but he did not think the result so clear that they should be warranted in abandoning the principles laid down by the Lightning Rod Conference.

M. de Fonvielle, of Paris, supported the views of Mr. Preece. He cited the example of Paris, where they had erected a sufficient number of lightning conductors, according to the received principles, and calamities from lightning were practically unknown. He suggested that the Eiffel Tower, which they were now building, and which would be raised to the height of a thousand feet, would furnish an unrivalled opportunity for experiments on lightning conductors.

Sir James Douglass, Chief Engineer to the Corporation of Trinity House, had a large experience with lighthouse towers. The lightning rods on these towers had been erected and maintained during the last fifty years entirely according to the advice of Faraday. They never had a serious accident; and such minor accidents as did occur from time to time were always traced to some defect in the conductor. They had now established a more rigid system of inspection, and he, for one, should feel perfectly safe in any tower where this system was carried out.

Mr. Symons, F.R.S., Secretary to the Meteorological Society, had taken part in a discussion on lightning conductors as long ago as 1859. It had been a hobby with him all his life to investigate the circumstances of every case he came across in which damage was done by lightning, and the general impression left by his investigations entirely coincided with the views just expressed by Sir James Douglass. He had been a member of the Lightning Rod Conference, and was the editor of their report; and he wished to enter his protest against the idea of rejecting all that had hitherto been done in connection with lightning conductors on the strength of mere laboratory experiments.

Professor Lodge, in reply, said he could perfectly understand the position of those who held that a lightning rod properly fitted up never failed to do its duty, because, whenever it failed, they said it was not properly fitted up. The great resource in such cases was to ascribe the failure to bad earth

contact. He thought a good earth contact was a very good thing, but he could not understand why such extraordinary importance should be attached to it. A lightning rod had two ends—an earth end and a sky end—and he did not see why good contact was more necessary at one end than at the other. If a few sharp points sticking out from the conductor were sufficient for a good sky contact, why were they not sufficient also for a good earth contact?

Besides, though a bad earth contact might explain why a certain amount of disruption should take place at the earth where the bad contact existed, he did not see how it accounted for the flash shooting off sideways half-way down the conductor. Again, what does a bad earth contact mean? If an electrical engineer finds a resistance of a hundred ohms, he will rightly pronounce the earth contact to be very bad indeed. But why should the lightning flash leave a conductor with a resistance of a hundred ohms in order to follow a line of non-conductors where it encounters a resistance of many thousand ohms?

He accepted the statement of Mr. Preece that his whole theory depended on the existence of oscillations in the lightning discharge; but there was good reason to believe they existed, because they were proved to exist in the discharge of a Leyden jar. Mr. Preece objected that an oscillating discharge could not produce magnetic effects, as a flash of lightning was known to do. He confessed he was unable to explain how an oscillating discharge produced such effects;^[43] but that it could produce them there was no doubt whatever, for the discharge of a Leyden jar produces magnetic effects, and we have ocular demonstration that the discharge of a Leyden jar is an oscillating discharge.

As to the assurances we had received from electrical engineers that a properly fitted lightning conductor never fails, he should like to ask them how the Hotel de Ville, in Brussels, had been set on fire by lightning on the 1st of last June. The system of lightning conductors on this building had been erected in accordance with the received theory, and had been held up by writers on the subject as the most perfect in Europe. Unless some explanation were forthcoming to account for its failure, we could no longer regard lightning conductors as a perfect security against danger.

The President of Section A, Professor Fitzgerald, in bringing the discussion to a close, observed that one result of this meeting would be to give a new interest to the phenomena of static electricity and its practical applications. He was inclined himself to think that the experiments of Professor Lodge were not quite analogous to the case of a flash of lightning. In comparing the discharge of a Leyden jar with a flash of lightning they should look for the analogy, not so much in the external discharge through a series of conductors, but rather, as Mr. Preece had observed, in the bursting of the glass between the two coatings of the jar. As regarded the oscillations in a Leyden jar discharge, he did not think such oscillations were at all necessary to account for the phenomena observed in the experiments. Many of the results which Professor Lodge seemed to think would require some millions of oscillations per second would be produced by a single discharge lasting for a millionth of a second. Improvements, perhaps, were possible in our present system of lightning conductors, but practical experience had shown, however we might reason on the matter, that, on the whole, lightning conductors had been a great protection to mankind from the dangers of lightning.

Summary.—I will now try to sum up the results of this interesting discussion, and state briefly the conclusions which, as it seems to me, may be deduced from it. First, I would remind my readers that a lightning rod has two functions to fulfill. Its first function is to promote a gradual, but rapid, discharge of electricity according as it is developed, and thus to prevent such an accumulation as would lead to a flash of lightning. Its second function is to convey the flash of lightning, when it does come, harmless to the earth. Now, the new views advanced by Professor Lodge in no way impugn the efficiency of lightning rods as regards their first function; and it is evident that the greater the number of lightning rods distributed over a given area, the more perfectly will this function be fulfilled. This is a point of great practical importance which seemed to me, in some degree, lost sight of during the progress of the discussion.

Secondly, it was practically admitted by the highest authorities that the experiments and reasoning of Professor Lodge afford good grounds for reconsidering the received theory of lightning conductors as regards their second function—that of carrying the lightning flash harmless to the earth. But there was undoubtedly a general feeling that it would be rash to set

aside, all at once, the received theory on the strength of laboratory experiments made under conditions widely different from those which actually exist in a lightning discharge. Experiments are wanted on a larger scale; and, if possible, experiments with lightning rods themselves.

Thirdly, the testimony of electrical engineers who have had large experience with lightning conductors seems almost unanimous that a lightning conductor erected and maintained in accordance with the conditions prescribed by the Lightning Rod Conference gives perfect protection. It was certainly unfortunate that the Hotel de Ville, in Brussels, which was reputed the best protected building in Europe, should have been damaged by lightning just two months before the discussion took place; but no certain conclusion can be drawn from this catastrophe until we know exactly the conditions under which it occurred.

So the matter stands, awaiting further investigation.

FOOTNOTES:

[41] See his Lectures, published in the *Electrician*, June 22, June 29, July 6, and July 13, 1888.

[42] See paper read at the meeting of the British Association, in Bath, 1888, published in the *Electrician*, page 607. September 14.

[43] See a very ingenious hypothesis, to account for this phenomenon, suggested by Professor Ewing in the *Electrician*, p. 712. October 5, 1888.

Transcriber's Notes

Errors and omissions in punctuation have been corrected.

[Page 11](#): "continuous inpression" changed to "continuous impression"

[Page 41](#): "full conconducting power" changed to "full conducting power"

[Page 44](#): "it base" changed to "its base"

[Page 58](#): "follow one an-another" changed to "follow one another"

*** END OF THE PROJECT GUTENBERG EBOOK LIGHTNING,
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